Detecting unknown arbuscular mycorrhizal fungi from linked long-read rDNA copies

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Abstract

1. Arbuscular mycorrhizal fungi (AMF) are named from environmental DNA by three systems built on different ribosomal DNA (rDNA) regions: MaarjAM virtual taxa (VTs) on the small-subunit gene (SSU), UNITE species hypotheses (SHs) on the internal transcribed spacer (ITS), and species clades on large-subunit (LSU) reference trees. The names rarely agree, and rDNA copies differ within one AMF genome, so one fungus can look like several taxa and a divergent copy like a new one. We asked how the names relate, whether they combine into one reliable call, and how well that call flags absent species.
2. We named 903 long-read rDNA copies spanning SSU, ITS and LSU, from 151 cultures of 37 species, with all three systems. We withheld each culture and species in turn, also removed withheld species from the LSU reference, tested 52 cultures from other collections against predictions fixed in advance, and checked divergent copies on trees and for artifacts and rRNA structure.
3. SHs mostly nest within LSU species, and these within VTs. VTs separated at most 19 of 37 species, whereas single cultures spanned 2–6 SHs (74 of 107 cultures). Copies with a shortened 5.8S gene resembled new lineages but share one deletion across isolates of several species: they are genomic paralogs, not artifacts. On trees, species formed clades; cultures rarely did. On the Krüger fragment, 97% identity split 4.1% of copies from their culture, against 24.4% on the full ITS at 98.5%. With a culture withheld, combined names kept the species, alone or in a group, for 745 of 755 copies. For withheld species also removed from the LSU reference, 11.6% of copies were still called known, 7.9% after an ITS distance rule (2.8% and 1.3% with the reference intact). All remaining errors fell within 4 species complexes (9 species) that the markers cannot separate. In cultures from other collections, the only false-known calls (4 of 102 sequences) were those predicted.
4. Novelty can be called from rDNA only where the markers separate species, and only from evidence independent of the database that defines what is known. Divergent copies should be recognized before taxa are counted, and inseparable species reported as groups.

Keywords. arbuscular mycorrhizal fungi; Glomeromycota; rDNA intragenomic variation; MaarjAM; UNITE; novelty detection; phylogenetic placement; long-read metabarcoding.

1 Introduction

Surveys of arbuscular mycorrhizal fungi (AMF) name what they find in three incompatible ways, so results from different markers cannot be pooled. All three read the ribosomal DNA (rDNA)

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unit, which holds the genes for the small-subunit (SSU, 18S), 5.8S and large-subunit (LSU, 28S) ribosomal RNAs, separated by two internal transcribed spacers, ITS1 and ITS2 (with the 5.8S gene between them, the ITS; Figure 1a). SSU studies report MaarjAM virtual taxa (VTs), groups of SSU sequences defined by phylogeny and at least 97% similarity [40, 39]. ITS studies report UNITE species hypotheses (SHs), clusters of ITS sequences at fixed distance thresholds [25, 34]. LSU studies place reads on curated reference trees and name them by the species clade they fall in [27, 11, 9, 10]. No agreed table maps one system onto another, and global occurrence databases such as GlobalAMFungi keep the datasets apart [50]. The names in the three databases were assigned independently, many reference sequences are unidentified environmental clones, and the taxonomy itself keeps moving: the family Glomeraceae has recently been divided into several families [8], and a phylogenetic classification of Glomeromycota has added new orders and families, some described from environmental DNA [46].

AMF genomes make the mapping harder. Their rDNA copies are not made uniform: copies within one individual differ [41], a single *Rhizophagus irregularis* genome carries only ten or eleven complete 45S rDNA units that differ in sequence and are not arranged in tandem [31], and copies within a culture can differ by several percent in the ITS [49, 45]. One genome can therefore look like several taxa in the ITS, and a divergent copy can look like an unknown lineage. The biology of AMF complicates naming further. A mycelium holds many nuclei, which in dikaryotic strains belong to two parental nucleotypes whose ratio shifts with the host plant [22]; nucleotype frequencies can change quickly after a host shift [2]; and *Rhizophagus irregularis* forms spores of two morphotypes, which has confused names in culture collections [23]. A genome-based species classification has recently been proposed for Glomeromycotina [7]. Any claim of a new AMF taxon therefore has to rule out divergent copies first. Rules for deciding when an ITS sequence is new have been calibrated for fungi in general [37], but not for AMF, in which three naming systems and variation within genomes interact.

Long reads that span the SSU, ITS and LSU link all three markers within a single read. Amplicons of about 2.5 kb were introduced to bring the three barcoding regions into one dataset [24], and a large collection of such copies from identified cultures now exists [45]. In this collection, every copy can be named by all three systems at once, and every culture serves as ground truth for its own copies. We used it to ask four questions:

1. How do VTs, SHs and LSU species clades map onto each other and onto cultured species?
2. Which kinds of divergent copies exist, are they genuine genomic copies, and how does each mislead naming?
3. Can the three names be combined into one reliable call?
4. How often is a species that no reference culture holds called a known species, and how much does the answer depend on what the external databases contain?

This work is part of amfsplit, a long-read AMF metabarcoding tool built on ITSxRust v0.3.0 [35]. We report the part that names each copy in the three systems and decides when a copy is new (Figure 1). Cutting reads into the regions of the three databases is described only as far as the deposited copies need; the tool's performance on environmental reads, and its correction of abundances for copy number, will be reported with the tool.

2 Materials and methods

All analyses read one fixed, versioned reference of culture copies, and every quantity learned from the cultures was refitted without the withheld cultures in each test (Section 2.6). Software versions, parameters and the scripts are given in Supplementary Methods S1–S8 and in the repository (Data and code availability).

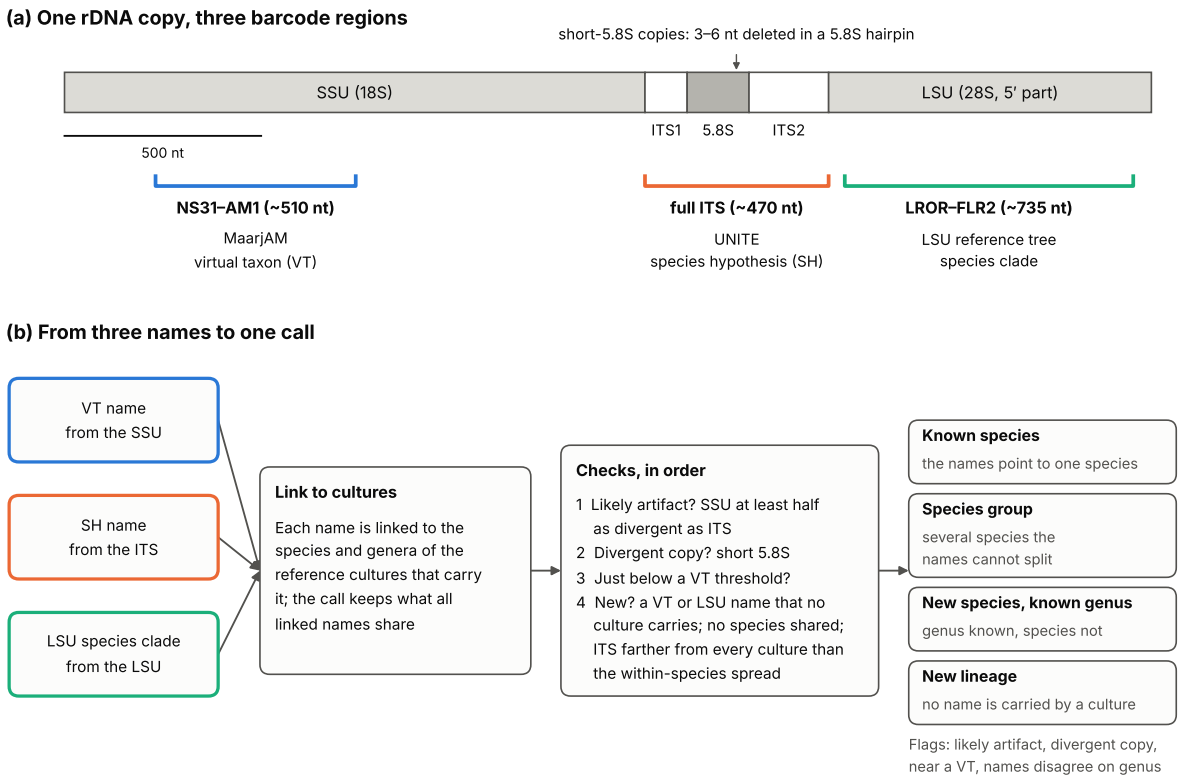


Figure 1: How a copy is named. (a) A linked rDNA copy at the median coordinates of the 903 culture copies (2,758 nt), with the three regions on which the naming systems are built: the NS31–AM1 fragment of the SSU for MaarjAM virtual taxa, the full ITS for UNITE species hypotheses, and the LROR–FLR2 fragment of the LSU for species clades on the LSU reference tree. The arrow marks the deletion carried by the short-5.8S copies (Section 3.3). (b) Each copy receives a name in each system; each name is linked to the reference cultures whose copies carry it, and the call keeps what all linked names share. Simpler explanations are then tested before a copy is called new (Section 2.4).

2.1 Culture copies and barcode regions

We used the 904 rDNA copies deposited by Stefani et al. [45] (GenBank PX207096–PX207274 and PX214440–PX215164). Each copy is a PacBio read of about 2.8 kb that spans most of the SSU, the whole ITS and the first part of the LSU (Figure 1a). Their study reports PacBio rDNA for 148 cultures of 44 species and three protein-coding genes for 143 cultures of 39 phylogenetic species; we used the rDNA copies and, only to examine species that rDNA cannot separate, the protein-coding genes (Section 4.6). The deposit carries 148 culture labels and 41 names (36 species and 5 named only to genus), the same names as in their culture tables. Three labels hold two lineages each (mixed cultures), which we counted as separate cultures, giving 151 cultures. We used the names as deposited, treating *Rhizoglomus* and *Rhizophagus*, both in use for this lineage, as one genus where genera are compared; the reorganization of the Glomerales [8] postdates most deposits, and a name that is outdated in a database is not an error of the marker.

In each copy, ITSxRust located the SSU, ITS1, 5.8S, ITS2 and LSU, and from each copy we cut, at the original primer sites, the fragment each database was built on: NS31–AM1 for MaarjAM [44, 17, 40], the full ITS (ITS1, 5.8S and ITS2) for UNITE, and LROR–FLR2 for the LSU reference [5, 47, 11]. Four further fragments used in AMF studies were cut the same way (Table S1; Methods S1). The trees and the structure check used the three regions that together

make up the whole copy: the SSU flank (all SSU sequence in the copy), the full ITS and the LSU flank (all LSU sequence).

The reference was built from the copies and a file of curation edits, each of which stated in advance what the data had to look like; a failed expectation stopped the build (Table S2; Methods S2). The final version (version 4) holds 903 copies of 151 cultures with 41 names: 37 named species and 4 names given only to genus. Within a culture, most copies share one main rDNA type. Copies whose 5.8S gene differed by more than 3% from that of the culture's medoid (the copy closest to all others of its culture) were set apart as *divergent* and split into two classes: *short-5.8S* copies, whose 5.8S is at least 3 bp shorter than that of the culture's main type, and *same-length* copies. All other copies are *main-type* copies.

The distance between two sequences was the number of differences (substitutions, insertions and deletions) in their best overlap, as a percentage of the shorter sequence (edlib [52]).

2.2 Naming each copy in the three systems

Each copy was searched with BLAST+ [6] against the database of each system, with the identity thresholds of that system (Table 1): its SSU flank against MaarjAM, whose references span the NS31–AM1 fragment (the gDAT snapshot of July 2020 [40]); its full ITS against UNITE 10.0 (the full UNITE+INSD set of 19 February 2025 [34, 1]); and its LROR–FLR2 fragment against the LSU reference of Delavaux et al. (V18 of May 2025 [11, 9, 10]). Hits to any copy of the culture reference were dropped, so no copy could take its name from itself or from a copy of its own culture. Instead of the best hit alone, each copy kept every name whose identity lay within a small margin of its best hit (its *candidate names*). The margin was the smallest at which the main-type copies of at least 99% of cultures share a name, and names that split the main-type copies of one culture were merged when the copies on either side lay within 1% of each other in the SSU (Table S3; Methods S3). For the LSU, copies were also placed on the V18 reference tree [3], and a placement within a clade that holds a single V18 species was read as that species' clade (the *LSU species clade*; Methods S4).

Table 1: The three naming systems. Each region of a copy is searched with BLAST+ against one external database per system; every search requires an e-value of at most 10^{-50}.

System	Region	BLAST task	Database	Identity	Coverage
VT	SSU flank	megablast	MaarjAM (gDAT, July 2020; 3,081 seq., 384 VT)	$\geq 97\%$	$\geq 95\%$
SH	full ITS	blastn	UNITE 10.0, UNITE+INSD (16,800 Glomeromycota seq., 7,563 SH)	$\geq 98.5\%$	$\geq 90\%$
LSU	LROR–FLR2	blastn	Delavaux V18 (460 AMF records, 234 species labels)	$\geq 97\%$	$\geq 90\%$

Coverage is of the shorter sequence, with reference Ns not counted; MaarjAM references span the NS31–AM1 fragment. Hits to any accession of the culture reference are dropped. The SH search was also run against the UNITE general release (2,048 Glomeromycota representatives).

2.3 Combining the three names

Each name was linked to the species and genera of the reference cultures whose copies carry it (Figure 1b). In each system, a copy's name was *carried* when at least one reference culture carries it and *not carried* when none does; a copy without a hit had no name in that system. The combined call kept the genera shared by all carried names and the species shared by the carried names that are linked to named species. The result was a species, a species group (several species that the names cannot split), a genus, *discordant* (the carried names share no genus) or unplaced.

2.4 Deciding when a copy is new

Before a copy was called new, simpler explanations were tested in a fixed order (Figure 1b; Methods S5):

1. **Likely artifact.** Against the nearest main-type copy of another culture, the SSU differs by at least 1% and by at least half as much as the full ITS. In functional rDNA the SSU changes far more slowly than the ITS (Section 3.4), whereas random errors, and copies freed from selection, change both at similar rates.

2. **Divergent copy.** The copy's 5.8S is at least 3 bp shorter than, and at least 3% different from, the main-type 5.8S of the nearest culture; or its best VT or SH is reached, in at least two other cultures, only by short-5.8S copies (a *paralog name*). Such copies are named by their VT alone, because their SSU is that of their genome. A same-length 5.8S that differs by 3% or more is only flagged.

3. **Near a VT threshold.** The copy has no VT name, but its best VT lies within one percentage point of the 97% threshold.

4. **New.** A species call was demoted to *new species in a known genus* when the VT or the LSU name is not carried, when the carried names share a genus but no species, or when the full ITS lies farther from every reference culture than the within-species spread (the 99th percentile of the ITS distance between cultures of one species; the *ITS distance rule*). A copy none of whose names is carried was called a *new lineage*, which has no genus. An SH that is not carried does not demote a call; SHs often split a single genome (Section 3.2).

2.5 Species complexes

Some named species cannot be told apart by any of the three markers, and we defined this from the reference alone. For each marker (full ITS, LROR–FLR2 and SSU flank), the *within-species spread* is the 99th percentile of the distance from a main-type copy to the nearest main-type copy of another culture of the same species: 6.0% in the ITS, 3.0% in the LSU and 0.8% in the SSU. Two species are *inseparable* when, on every marker, their nearest copies lie no further apart than this spread, so that a copy of one falls inside the spread of the other. Inseparable pairs that share a species were joined into a *complex*, and a complex is reported as one unit, a species group, wherever a species call is made. The rule uses no test result and adds no threshold beyond the percentile that the ITS distance rule already uses. Because that percentile was chosen after the distances in these cultures had been seen, the ITS distance rule and the complexes were also recomputed together at the 95th and 97.5th percentiles (Methods S6).

2.6 Tests with withheld cultures and species

We tested the calls by withholding parts of the reference and asking how the withheld copies were named. In each test round, everything learned from the cultures (merged names, margins, the links from names to species, the comparisons of the divergent-copy test, the paralog names and the limit of the ITS distance rule) was refitted on the remaining cultures only (Methods S6).

Withheld cultures. In 148 rounds, one culture label was withheld, together with both lineages where a label holds a mixed culture. The question was whether the main-type copies of species still held by another culture kept a call that contains their species.

Withheld species. In 50 rounds, one named species was withheld with all its cultures (37 rounds), or one culture not named to species (13 rounds). A *false-known call* is a call of a known species or species group for a copy whose species has no culture left in the reference.

Withheld from the LSU reference as well. In the tests above the three databases stayed intact, so a withheld species could still have its own VT, SH or LSU clade. In the *LSU-removal test*, each withheld species was also removed from the LSU reference: the reference alignment was rebuilt without its records, the model refitted and all copies placed again. MaarjAM and UNITE were not edited, so a withheld species kept any VT or SH it has there; this matters because a VT that is not carried demotes a call and an SH does not. The ITS distance rule was added after two species, *Glomus chinense* and *Glomus rugosae*, had been run in this test, and these are reported as development species. The 34 other species whose LSU records could be removed (the *test species*) were run after the rule had been fixed; *Ambispora callosa*, which the LSU reference does not hold by name, is reported apart. In 15 further rounds, every culture of a genus and the genus's LSU records were withheld together.

These cultures were used to develop the methods, so every result from them is a development estimate. The independent test is described next.

2.7 Cultures from other collections

LongGloDB v1 (July 2026) [14] holds 1,122 curated SSU–ITS–LSU amplicon sequence variants (ASVs) from AMF culture collections. We left out the 810 ASVs that carry the laboratory codes of the Stefani et al. cultures (STFP and STFI): 784 of them are identical in full ITS to a copy in our reference, so they are the same cultures processed again. A further 4 cultures share a culture code with a Stefani label (3 of them under another species name) and were left out too. The test set holds the remaining 280 ASVs of 52 cultures from three other laboratories: 164 ASVs of 22 cultures of species the reference holds, 59 of species it does not hold in genera it does, 43 of genera it does not hold, and 14 named only to genus. The ASVs of absent species come from 26 cultures of 19 species.

Before any test sequence was searched, placed or scored, we applied the complex rule (Section 2.5) to the test ASVs and fixed its predictions, together with a protocol and two amendments that set the outcomes and analyses, by recording their checksums, digital fingerprints that show any later change (Methods S7). A test ASV can be wrongly called a known species only if it lies inside the within-species spread of that species on all three markers, so a false-known call outside the predicted set would contradict the result of Section 3.7. The rules were used unchanged and learned only from the 151 reference cultures, so no test sequence influenced them. The test sequences were cut like the deposited copies and named against the same database releases, first with the LSU reference intact and then with each of the 19 absent species removed from it in its own round, as in the LSU-removal test.

The test amplicons begin about 60 bp inside the NS31–AM1 fragment, so they rarely cover the 95% of a MaarjAM reference that the VT search requires, and only 33 of the 280 ASVs had a VT hit that passed. The main analysis applied the rules as fixed. A second analysis, also fixed before the run, counted VT coverage over the part of each MaarjAM reference that the ASV spans and changed nothing else.

2.8 Trees, artifact check and structure check

These checks were fixed in a protocol, and its checksum recorded, before they were run; the structure check was added in a dated addendum that was likewise fixed before its results were seen (Methods S8).

Trees. For each order, identical sequences were collapsed and the copies aligned with MAFFT L-INS-i [21], separately for the SSU flank, the full ITS and the LSU flank, and for the three joined end to end (the *linked copy*). Maximum-likelihood trees were inferred with IQ-TREE 2 [32] with ModelFinder [20], 1,000 ultrafast bootstraps [18] and 1,000 SH-aLRT replicates [16]. A culture, named species or complex was *monophyletic* when one branch of the unrooted tree separated exactly its copies from all others, and *strongly non-monophyletic* when a branch that

contradicts it had SH-aLRT ≥ 80 and ultrafast bootstrap ≥ 95. Each group was tested with all copies and with the divergent copies removed.

Artifact check. Each divergent copy was aligned to the most similar main-type copy of its own culture. A deletion was *long* when it reached, in its region, the 99th percentile of the longest deletion between main-type copies of one culture, with a floor of 3 nt: 3 nt in the SSU, 5.8S and LSU and 5 nt in ITS1 and ITS2. A divergent copy *recurred* when a copy within 1.0% of it in full ITS occurred under another culture label, that is, in another isolate. The verdicts were fixed in advance: a recurring copy without long deletions is a paralog; a recurring copy with them is a recurrent deleted copy (genomic, possibly a pseudogene); a single copy with them is a probable artifact; and a single copy without them is unresolved.

Structure check. To ask whether divergent copies keep the structure of functional rRNA, the SSU, 5.8S and LSU of every copy were aligned with Infernal [33] to the Rfam models of eukaryotic SSU rRNA (RF01960), 5.8S rRNA (RF00002) and eukaryotic LSU rRNA (RF02543) [38], which describe both the sequence and the secondary structure of each RNA family. Each base pair of the consensus structure that a copy covers was *broken* when it was not a Watson–Crick or G·U pair, or when one side was deleted. A copy was *degraded* in a region when its share of broken pairs exceeded the 99th percentile among main-type copies, and *irregular* when its substitutions or insertion and deletion events, counted against the most similar main-type copy of its culture, exceeded the 99th percentile among main-type copies. The verdict for the short-5.8S class was fixed in advance: *structurally intact* if at most 10% of its copies were degraded in each region and its shared deletion removed only unpaired positions of the consensus structure; *structurally degraded*, as expected of pseudogenes, if at least 50% were degraded in some region or the shared deletion removed paired positions; and *mixed* otherwise. The ITS2 of every copy was folded with RNAfold [30], for description only.

2.9 Distances at each level of relatedness

To explore similarity thresholds without fixing any rule, we measured, for every main-type copy, the distance to its nearest main-type copy in four classes: the same culture, another culture of the same species, another species of the same genus, and another genus of the same order. We did this for the NS31–AM1 fragment, on which VTs are defined, the whole SSU flank, the full ITS and LROR–FLR2, and we also measured the distances between the medoid copies of every pair of cultures within an order. The same distances were measured on two longer regions, the Krüger fragment and the whole copy. We counted what the short type does to clusters and to the tree of each marker, and tested a sequence motif that flags it (Methods S8).

3 Results

3.1 The culture copies

ITSxRust located the regions in all 904 deposited copies, and every fragment whose primers lie inside a copy was recovered (Table S1). The fragments are close in length to the amplicons behind the databases: NS31–AM1 has a median length of 508 bp without its primers and about 550 bp with them, the size of the fragment behind MaarjAM virtual taxa. The 28 copies without an NS31–AM1 fragment are exactly the Paraglomerales copies, in which the AM1 primer does not bind.

Curation changed little (Table S2). One copy was dropped because its SSU and ITS differ from the other copies of its culture at the same rate, the signature of an error or a pseudogene. One culture held a second *Paraglomus* lineage, and two *Septoglomus* cultures, SO43BL and Blasz_17_2277A, carry the same two lineages with the majority reversed; these were split. The two *Septoglomus* lineages differ as species do (ITS1 11–12%, 5.8S 5.7%, LSU 1.6%). Two *Rhizophagus irregularis* cultures were kept under their name although their SSU falls in VTX00113,

a VT that also holds *[Rhizoglomus] vesiculiferum*: their ITS and LSU are closer to *Rhizophagus irregularis* (in T3_IVT34A, the ITS2 lies a median 3.3% from the nearest *Rhizophagus irregularis* copy and 6.4% from the nearest *[Rhizoglomus] vesiculiferum* copy). In two cultures the medoid copy fell among the short-5.8S copies, which inverted the classes; we set them by 5.8S length. In one of these, SD5_IVT32, the corrected main type matches the V18 record of *Rhizophagus proliferus* at 98.5–99.9% identity, while its short-5.8S copies are 91.8% from anything in V18. The final reference holds 903 copies of 151 cultures of 37 named species, with 51 divergent copies: 43 short-5.8S copies in 24 cultures and 8 same-length copies in 5 cultures.

3.2 The three naming systems nest

The SSU gave a VT name to 895 of the 903 copies. Counting each culture by its most common VT, 3 of 37 named species span more than one VT, and 8 of 30 VTs hold more than one species; VTX00199 alone holds four *Glomus* species. The nearest LSU reference gives a culture its own species for 105 of the 135 named cultures whose species V18 holds (78%; 24 get a congener and 6 no name), and the right genus for all that get a name. V18 itself, however, holds two records for nearly every species (460 records under 234 species labels: 215 labels with two records, five with three or four, and 14 with one); 45 of the 220 labels with two or more records hold two records that are less than 97% identical, and when each record was left out in turn, only 72% of records returned their own species.

UNITE+INSD, which holds all public sequences, named 592 copies (66%), against 383 (42%) for the general release, which keeps one representative per SH. An SH is finer than a genome: the main-type copies of 74 of the 107 cultures with two or more SH names fall in 2–6 SHs. Counting each culture by its most common SH, 0 of 70 SHs hold two named species, and VTs split 0 of 77 SHs; once SHs are merged into one group per genome, one group is shared by *Glomus macrocarpum* and *Glomus rugosae*. The LSU splits 1 of 73 SHs (into the V18 species *Diversispora varaderana* and *Diversispora bareai*), and VTs split 4 of 40 LSU names. So SHs mostly sit inside LSU species, which mostly sit inside VTs.

Within a culture the SSU hardly varies (median 0.07%, 95th percentile 0.54%, 99th percentile 0.95%). Merging two pairs of VTs, VTX00113 with VTX00115 and VTX00125 with VTX00222, gives every culture one consistent set of VT names (Table S3), and the margin needed to absorb the SSU differences within cultures (0.6 points) costs three species of resolution. SHs merged into one group per genome separate 32 of the 37 species, against 16 for VTs; this is an upper bound, because the merges were learned on the same cultures.

3.3 Divergent copies are genomic paralogs

The two divergent classes mislead each marker differently (Table 2). Short-5.8S copies are well represented in public data, as expected for the variable, non-tandem rDNA units of a single genome [31]: their median best SH hit (99.8%) is closer than that of main-type copies (99.1%). Divergent copies reach 14 SHs, of which 11 are reached by no main-type copy (paralog SHs, 4 of them under placeholder names) and 3 by one main-type copy each (Table 3). The short copies of *Septoglomus viscosum* are the only culture copies that reach VTX00409, about two points clear of VTX00063. In *Gigaspora rosea*, four cultures carry the same three or four rDNA types, identical across cultures and up to 6.2% apart in the LSU, each in its own SH named *Gigaspora rosea*. These types are not confined to one isolate held in several collections: in LongGloDB (Section 3.8), INVAM cultures under other accession codes (BR155B, FL105 and KS885; FL105 is INVAM's reference culture for the species) carry ASVs within 0.25% of them in full ITS.

They are not artifacts. Under the rule fixed in advance (Table 4), 3 same-length copies are paralogs (the *Gigaspora rosea* types, which match main-type copies of other cultures), 22 short-5.8S copies are recurrent deleted copies, and 21 short-5.8S and 3 same-length copies are

Table 2: The two divergent copy classes mislead each marker differently.

	Short 5.8S	Same length
Copies (cultures)	43 (24)	8 (5)
VT: same as the culture's main type / other / no call	35 / 7 / 1	8 / 0 / 0
LSU: same species / another species / no call	5 / 1 / 31	2 / 5 / 1
LSU best-hit identity, median (main type 98.9%)	94.9%	97.2%
SH: same / other SH / no call	0 / 28 / 7	0 / 4 / 1
SH best-hit identity, median (main type 99.1%)	99.8%	98.6%

Each divergent copy compared with the most common name of its culture's main-type copies (culture reference version 4). Copies in cultures without a main-type name in a system are left out of that system's row (short 5.8S: VT 0, LSU 6, SH 8; same length: VT 0, LSU 0, SH 3).

Table 3: Paralog names: SHs and a VT reached by divergent copies and by at most one main-type copy.

Name	Taxon of the best hit	Divergent copies	Main	Identity
SH1337353.10FU	*Rhizophagus irregularis*	*Rhizophagus irregularis* 10 (7 cultures)	1	100.0%
SH1364117.10FU	Glomeromycota sp.	*[Rhizoglomus] vesiculiferum* 7 (2 cultures)	0	98.5–99.8%
SH1081549.10FU	*Rhizophagus* sp.	*Rhizophagus prolifer* 4 (1 culture)	0	99.3–99.8%
SH1081557.10FU	*Rhizoglomus intraradices*	*Rhizophagus intraradices* 4 (3 cultures)	0	100.0%
SH1076762.10FU	*Gigaspora margarita*	*Gigaspora rosea* 2 (1 culture)	1	99.8%
SH1337340.10FU	*Rhizophagus* sp.	*Rhizophagus* sp. 1; *Rhizophagus irregularis* 1 (2 cultures)	0	100.0%
SH1337458.10FU	*Rhizophagus irregularis*	*Rhizophagus silesianus* 2 (1 culture)	0	100.0%
SH1389224.10FU	Glomeraceae sp.	*Septoglomus viscosum* 2 (1 culture)	0	100.0%
VTX00409	environmental (MaarjAM)	*Septoglomus viscosum* 3 (2 cultures)	0	98.5–99.4%

Main: main-type copies of any culture with that name. 6 further names take one divergent copy each (SH1076712, SH1196937, SH1337247, SH1337463, SH1337480, SH1364178). Taxa are those of the best UNITE record; the VT row is MaarjAM's.

probable artifacts. The rule counts a type as recurring only when its whole ITS recurs, and many short-5.8S copies fail that test because their spacers differ. Their deletion does not. In a reading made after the check, every one of the 43 short-5.8S copies, the 21 probable artifacts included, carries a 5.8S deletion of the same length, starting within 2 nt, as a copy under another culture label. In *Rhizophagus* and *[Rhizoglomus]*, all 40 short copies share one 6-nt deletion, found in 23 cultures of five named species and *Rhizophagus* sp., and so does one copy filed as main type (PX214776.1, *Rhizophagus irregularis* 4569_1185B, the main-type copy in SH1337353; Table 3). A deletion shared across species and isolates is not a PCR or sequencing error, so the short-5.8S copies form a genomic paralog family. The same-length probable artifacts rest on ITS2 deletions of 8–9 nt; with long deletions read as at least 10 nt, their number falls to 0.

Their 5.8S has lost part of one hairpin. The SSU of short-5.8S copies kept its secondary structure: 3 of the 43 copies were degraded, and their share of broken pairs did not differ from that of the main-type copies of their culture (median 10.0% against 10.6% among main-type copies; paired Wilcoxon test, $P = 0.17$). Their LSU carries far more substitutions than main-type copies (median 4.62 against 0.12 per 100 aligned positions), but most of these keep the base pairing: 12 copies (28%) were degraded, and the share of broken pairs differed little (7.6% against 8.5%; $P = 0.07$). The 5.8S differs most: 25 of the 43 copies were irregular, and broken

Table 4: Divergent copies by whether the same copy type occurs in other cultures (recurring) and whether it has long deletions.

	Recurring		Single	
Class (copies)	intact	deleted	intact	deleted
Short 5.8S (43)	0	22	0	21
Same length (8)	3	0	2	3

Recurring: a copy within 1.0% in full ITS occurs under another culture label. *Deleted*: compared with the most similar main-type copy of the same culture, the copy lacks at least 3 nt in a row in the SSU, 5.8S or LSU, or 5 nt in ITS1 or ITS2 (longer than 99% of such gaps between main-type copies). The four cells correspond to the verdicts fixed before the check: paralog, recurrent deleted copy, unresolved and probable artifact.

pairs rose from 4.0% to 16.0% (median paired difference +12.0 points; $P < 10^{-8}$). In all 40 short copies of *Rhizophagus* and *[Rhizoglomus]*, the deletion removes three of the nine base pairs of a hairpin near the 3′ end of the 5.8S (consensus positions 121–130 of the Rfam model), whereas in the 3 *Septoglomus viscosum* copies it removes only unpaired positions. Under the rule fixed in advance, the short-5.8S class is therefore structurally degraded, as pseudogenes would be. The hairpin is shortened rather than destroyed, however. The remaining six-pair stem is made only of Watson–Crick pairs in 36 of the 40 copies, and its new loop is UUCG in 25, UUUG in 10 and CUCG in 5; a UUCG loop closed by a C–G pair, as in most of these copies, forms an unusually stable hairpin [48]. The main-type copies with the same shortened hairpin, PX214776.1 and PX214775.1 (*Glomus* sp. 5089_2351A), were among those flagged by the artifact check. The same-length copies were structurally intact: none was degraded in any region. In ITS2, the predicted folds of short copies differed from those of their culture's main type more than main-type copies differ from each other (median base-pair distance 0.16 against 0.00 per nucleotide), with similar folding energies (-0.23 against -0.26 kcal mol^{-1} per nucleotide).

On trees, short copies group across species. Most short copies of *Rhizophagus* and *[Rhizoglomus]* do not sit with the main type of their own culture (Figure 2): 39 of the 40 fall in 4 clades made only of short copies, 3 of which join two taxa (36 copies). The groups of short copies, with SH-aLRT and ultrafast bootstrap support, are: *Rhizophagus irregularis* with *[Rhizoglomus] vesiculiferum* (26 copies and 1 copy filed as main type; 100/92); *Rhizophagus intraradices* with *Rhizophagus prolifer* (8 copies; 100/99); *Rhizophagus silesianus* (3 copies; 100/100); *Rhizophagus irregularis* with *Rhizophagus* sp. (2 copies; 100/100); *Rhizophagus irregularis* (1 copy). The short copies of *Rhizophagus intraradices* and *Rhizophagus prolifer*, two species that are not a complex, are closer to each other than to the main type of either species. The short type is therefore most simply explained as a paralog that arose before these species separated and has been kept in their genomes since.

The ITS and LSU carry the paralog; the SSU does not. Counted by their 5.8S sequence, the Glomerales hold 44 short copies: the 43 of the divergent class and PX214776.1. On the SSU flank they show no history of their own: 21 are identical to a main-type copy of their species, and all but one of the rest, a copy of *Rhizophagus* sp., sit next to main-type copies of their own species. On the full ITS, 43 fall in clades made only of short copies, 3 of which join two taxa (37 copies). On the LSU flank, 42 do, but most of these clades hold one species, and 2 join two taxa (10 copies). The markers then disagree: the 8 short copies of *Rhizophagus intraradices* and

Rhizophagus prolifer (99/100) are sister to *Sclerocystis sinuosa* on the LSU tree (100/100), but to the main type of their own species on the full-ITS and linked trees (Supplementary Data S1).

They raise ITS species counts. Single-linkage clusters of all copies of the 37 named species at 98.5% identity on the full ITS number 240 (Table S7). Main-type copies alone give 220, so most of the excess over the number of species comes from ordinary variation within genomes; the divergent copies add 18 clusters made only of short copies and 2 made only of same-length copies. Each of the 6 named species with the short type gains 1 to 7 such clusters, none of them shared with another species. On the Krüger fragment at 97%, the count falls to 56 (46 from main-type copies), 8 of them short-only.

A short motif marks them in *Rhizophagus*. The shortened hairpin has a sequence of its own: where the main type reads CACTCTCTGG(T/C)AACCCGGGGAGTATG, the short type reads CACTCTCTT(T/C)GGAGAGTATG. The 13-nt motif YTYGGRGARTATG, written from these copies, is in the 5.8S of all 41 short copies of *Rhizophagus* and *[Rhizoglomus]* and of none of their other 113 copies, and nowhere else in any copy. It misses the 3 short copies of *Septoglomus viscosum*, whose deletion lies elsewhere, and it is not unique to the paralog: it is the normal sequence of all 29 copies of *Oehlia diaphana*, and it is also in PX214775.1. It starts at position 121–122 of the 5.8S, downstream of where the ITS2 primers fITS7 and ITS3 bind (from positions 84–85 and 31), so ITS2 amplicons made with either primer carry it (Methods S8).

3.4 On trees, species form clades but cultures do not

Cultures were rarely monophyletic: 12%–20% of the 139 cultures with two or more copies, depending on the region and on whether divergent copies were kept (Table 5). A culture that is not monophyletic typically has most of its copies outside any clade made only of its own copies (median 67% on the linked copy, main type), mixed with copies of other cultures of the same species (Figure 2), as expected if the rDNA variants of a genome are older than the isolates that carry them. Named species mostly are clades: 28, 24 and 28 of the 37 species on the SSU flank, full ITS and LSU flank, and 31 on the linked copy. Removing the divergent copies made 4 more species monophyletic on the linked copy (*Rhizophagus intraradices*, *Rhizophagus irregularis*, *Rhizophagus prolifer* and *[Rhizoglomus] vesiculiferum*), 35 of 37 in all. The two that remain, *Diversispora aestuarii* and *Glomus rugosae* (only the latter strongly), belong to species complexes, and all four complexes are monophyletic as units (4/4).

Table 5: Cultures, named species and species complexes that form a single clade (are monophyletic) on trees of the culture copies.

	Cultures		Named species		Complexes	
Tree	all	main	all	main	all	main
SSU flank	17/139	19/138	28/37	28/37	3/4	4/4
Full ITS	18/139	24/138	24/37	29/37	2/4	3/4
LSU flank	19/139	22/138	28/37	30/37	3/4	4/4
Linked copy	23/139	28/138	31/37	35/37	3/4	4/4

Monophyletic groups / groups tested; a group is tested when it has two or more copies. *all*: all copies; *main*: the same trees with the divergent copies removed. One tree per order and region (maximum likelihood, IQ-TREE); the linked copy joins the three regions. Alignments and trees are in Supplementary Data S1.

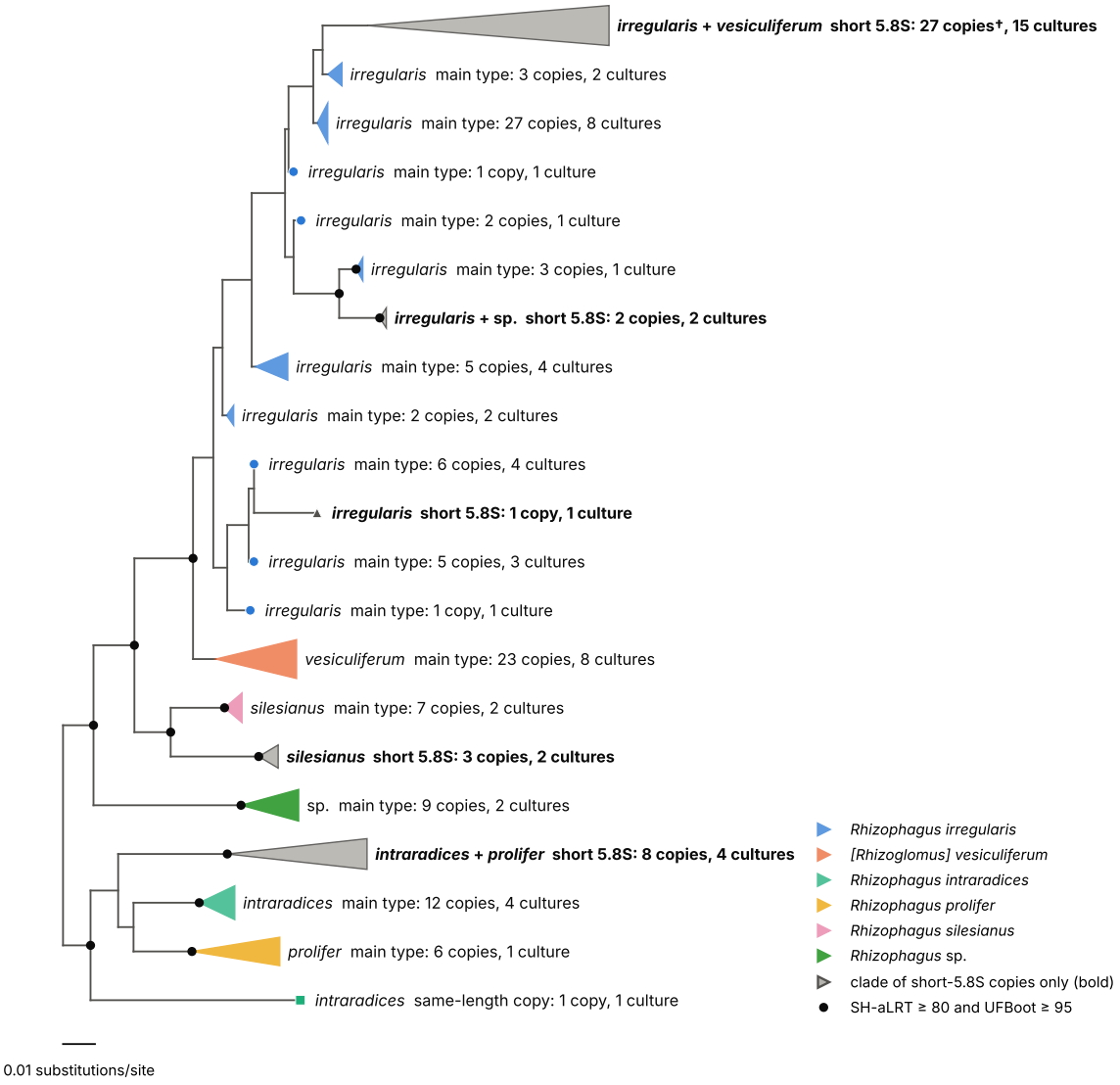


Figure 2: Short-5.8S copies group across species. The *Rhizophagus* clade, including *[Rhizoglomus] vesiculiferum*, of the maximum-likelihood tree of the linked copy (SSU flank, full ITS and LSU flank) of the Glomerales, rooted on *Funneliformis* (not shown). Clades made only of main-type copies of one species (colored), and clades made only of short-5.8S copies (grey, bold labels), are collapsed into triangles whose length is the depth of the clade; labels give the species epithet, the copy class and the numbers of copies and cultures. Black dots mark branches with SH-aLRT ≥ 80 and ultrafast bootstrap ≥ 95. †One of the 27 copies was filed as main type (PX214776.1, *Rhizophagus irregularis* 4569_1185B) but carries the deletion of the short copies. The uncollapsed tree (154 copies, 103 distinct sequences) is Figure S3, and all trees are in Supplementary Data S1.

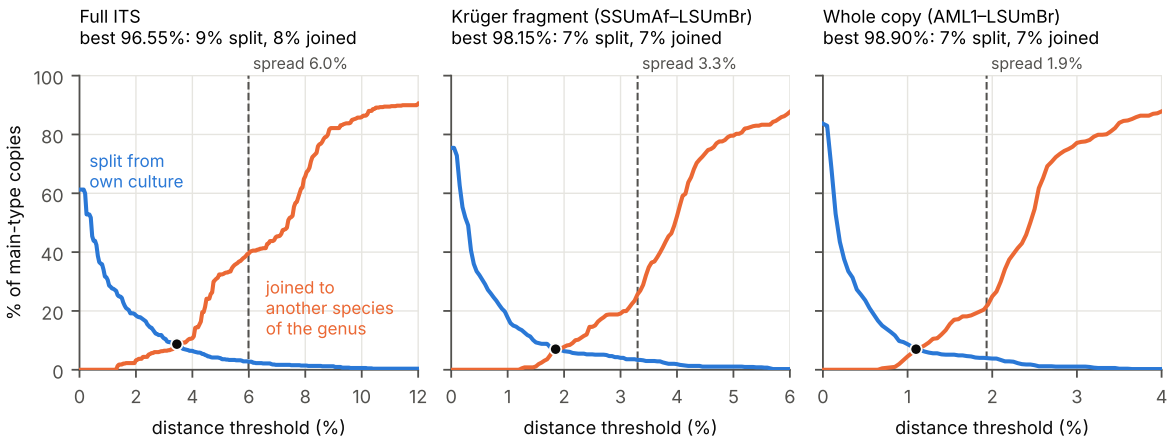


Figure 3: Single identity cut-offs on longer regions. For each cut-off, the share of main-type copies more than the cut-off from every other copy of their culture (blue) and with a copy of another species of their genus within it (orange), on the full ITS, the Krüger fragment (SSUmAf site to LSUmBr) and the whole copy (AML1 to LSUmBr). Dashed lines, the within-species spread; dots, the cut-off that minimizes the larger of the two shares, given above each panel. The four complexes count as separate species here; leaving their partners out, the share joined at the within-species spread falls to 11.8% (full ITS), 2.1% (Krüger fragment) and 1.9% (whole copy; Methods S8).

Distances tell the same story (Figure S1). A main-type copy lies about as close to its nearest culture-mate as to the nearest copy of another culture of its species (medians 0.41% and 0.66% in full ITS, 0.14% and 0.14% in the LSU). Congeners overlap this range on every region: 39.5% of main-type copies have a copy of another species of their genus within the within-species spread of the full ITS, 54.5% within that of the LSU and 80.1% within that of the SSU flank. Other genera stay outside it on every region except the NS31–AM1 fragment (13.5% of copies). The database thresholds fall at different points on this scale. On the NS31–AM1 fragment, 97% identity, the VT threshold, joins 98.7% of main-type copies with a congener and 15.2% with another genus (121 copies, all but 4 in *Cetraspora*, *Gigaspora* and *Dentiscutata*). On the full ITS, 98.5% identity, the SH threshold, lies well inside the within-species spread: 24.4% of main-type copies are more than 1.5% from every other main-type copy of their own culture, which helps explain why SHs split cultures. On the LSU, the 97% threshold of V18 coincides with the within-species spread (3.0%). Between pairs of cultures (Figure S2), the SSU-flank distance was a median of 0.14 times the ITS distance (pairs at least 2% apart in ITS). Only 6 of the 5,196 pairs lay above half the ITS distance, all of them cultures of one species with SSU distances of at most 0.48%, so no pair of cultures met the artifact test (SSU at least 1% and at least half the ITS distance).

Taken together, the regions of the Krüger fragment (from the SSUmAf site to LSUmBr; median 1,577 nt) keep genomes and species apart better than the full ITS alone (Figure 3). At 97% identity, 4.1% of main-type copies lie more than 3% from every other copy of their culture, against 24.4% for the full ITS at 98.5%, and outside the four complexes 0.3% have a congener within 3%. The complexes stay joined at any cut-off that keeps genomes together: counted as separate species, they leave the best single cut-off (98.15%) splitting 6.7% of copies from their culture and joining 7.0% to another species. The whole copy (median 2,758 nt) behaves the same way at a higher identity (best cut-off 98.90%: 7.0% split, 7.0% joined).

3.5 Combining the three names

Combining the names recovered what no single system reaches. With a culture withheld, the exact species of the 755 main-type copies of species held by another culture came from the VT alone for 370 (49%), from the SH alone for 343 (45%), from the LSU alone for 351 (46%) and from the combined call for 607 (80%); the checks for new copies then demoted 4 of these, leaving 603 (Table 6). Placing the LSU on the reference tree raised exact calls to 701 and cut false-known calls for withheld species from 26.8% to 8.8%, and the ITS distance rule cut them again to 2.2% (Figure 4a); averaged over species or over cultures, the final rate was 1.4% and 1.9%. The limit of the ITS distance rule was 6.0% in 139 of 148 rounds. The rule demoted 6 copies of known species (4 from *Glomus chinense* 4036_906B, 1 from *Gigaspora rosea* CU131_1018E, 1 from *Microdominikia litorea* 5274); the *Microdominikia litorea* copy lies 12.6% from the species' other culture. The 18 remaining false-known calls belong to two complexes: *Diversispora varaderana* with *Diversispora aestuarii* (9 copies), which no marker separates, and *Glomus chinense* with *Glomus rugosae* (9 copies), whose ITS ranges overlap.

Of the 42 divergent copies whose species is held by another culture, 35 were recognized as divergent, 3 same-length copies were called their own species, and 4 were wrongly called a new species. Short-5.8S copies were recognized from their own sequence in 36 of 43 cases with their culture withheld, and in 29 with their species withheld. Placement has a cost: 146 withheld copies lost a call of 'new species in a known genus' and became 'new lineage', which carries no genus.

Table 6: Calls for withheld cultures and species, with the external databases intact.

LSU naming	Known kept (of 755)	Exact species	False known species withheld (of 803)	False known single-culture species (of 48)
Nearest LSU reference	748	603	215 (26.8%)	3
LSU species clade (placement)	751	701	71 (8.8%)	0
Placement + ITS distance rule	745	697	18 (2.2%)	0

Known kept: main-type copies of species held by another remaining culture that receive a known species or species-group call containing their species (148 rounds, one culture withheld in each). *False known*: known calls for copies whose species has no remaining culture: in 50 rounds with one species withheld in each, and, in the rounds with one culture withheld, for the main-type copies of species with a single culture. Averaged over species and over cultures, the final false-known rate with a species withheld is 1.4% and 1.9%.

3.6 Removing the withheld species from the LSU reference

With the full LSU reference, copies of *Glomus rugosae* fall in the reference's own *Glomus rugosae* clade, which no remaining culture holds, and are flagged as new. With the species removed from the reference, they fall in the *Glomus chinense* and *Glomus macrocarpum* clades and are called known (Table 7; Figure 4b). The ITS distance rule does not use the LSU reference, but it rescued only 13 of the 75 copies, because the ITS of *Glomus rugosae* lies 1.4–7.5% (median 4.7%) from its relatives, within the ordinary spread of one species.

Pooled over the 34 test species (673 copies), false-known calls numbered 19 with the LSU reference intact, 78 with the species removed from it, and 53 with the ITS distance rule as well (Figure 4c; per species in Table S4); averaged over species, the rates were 20.6% and 11.7%. The 53 errors left after the ITS rule fall in 6 species pairs: *Diversispora aestuarii* called *Diversispora varaderana* (20), *Rhizophagus irregularis* called *[Rhizoglomus] vesiculiferum* (18), *Diversispora arenaria* called *Diversispora jakucsiae* (4), *Diversispora jakucsiae* called *Diversispora arenaria* (4), *Glomus macrocarpum* called *Glomus rugosae* (4), *Diversispora varaderana* called *Diversispora aestuarii* (3). Of the 673 withheld copies, 272 were called a new species in a known genus (195 because the LSU name was not carried, 34 because the names shared no species, 25 by

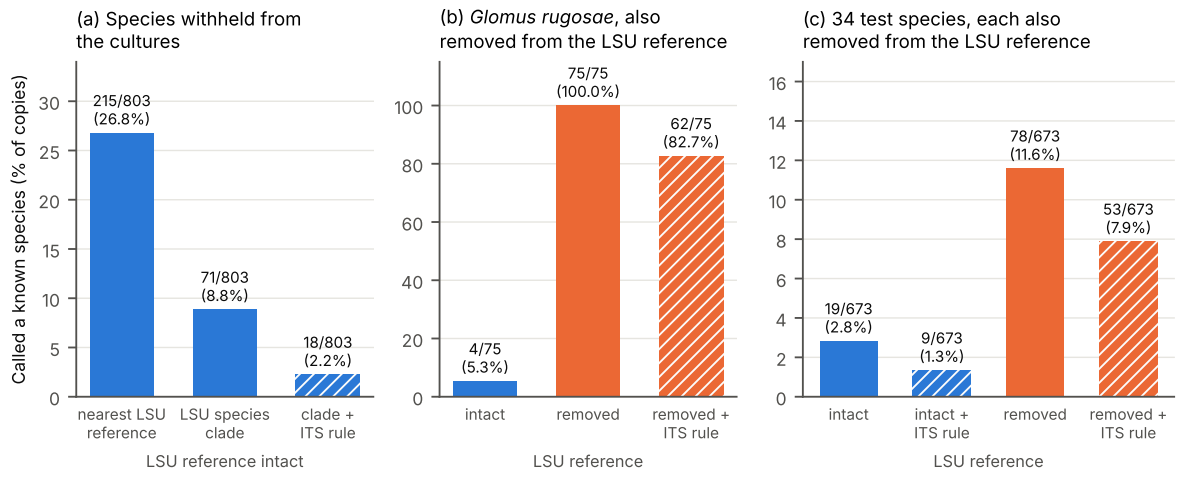


Figure 4: False-known calls: copies called a known species although no remaining culture holds their species. (a) Species withheld from the cultures, with the LSU reference intact, by how the LSU was used. (b) *Glomus rugosae*, a development species, also removed from the LSU reference: the signal of novelty that LSU placement gave came from the LSU reference itself. (c) The 34 test species, each also removed from the LSU reference, pooled. Blue, LSU reference intact; orange, species removed from it; hatched, with the ITS distance rule.

the ITS distance rule, and 18 because the VT was not carried), 347 a new lineage, 53 a known species or group, and 1 a divergent copy. The genus was named for 326 copies and was right for 326 of them, but that is precision. Of the 487 copies whose genus another culture holds, the genus was kept for 326 (66.9%), and 161 were called a new lineage although their genus was known; the other 186 copies belong to genera with no other culture and were all called a new lineage. The VT thus contributes little to the detection of new species (2.7% of copies), and only when the withheld species has a VT of its own. *Ambispora callosa*, which the LSU reference does not hold by name, was called a known species for 20/20 copies without the ITS rule and 0/20 with it, because its ITS lies about 10% from its closest relative.

Table 7: Withheld species also removed from the LSU reference.

Withheld species	LSU reference	False known	with ITS rule
Glomus chinense†	intact	28/35	5/35
Glomus chinense†	species removed	28/35	5/35
Glomus rugosae†	intact	4/75	4/75
Glomus rugosae†	species removed	75/75	62/75
34 test species	intact	19/673 (2.8%)	9/673 (1.3%)
34 test species	species removed	78/673 (11.6%)	53/673 (7.9%)

False-known calls among the main-type copies of the withheld species. † Development species: these two informed the ITS distance rule. *Intact*: copies placed on the full LSU reference (V18); *species removed*: V18 realigned without the species' records, the model refitted and all copies placed again. The test species pool the species whose V18 records could be removed; *Ambispora callosa*, which V18 does not hold by label, is reported apart.

With a whole genus withheld from the cultures and from the LSU reference (15 rounds, 803 main-type copies of named species), no copy was called a known species, with or without the ITS rule, as expected, because no remaining culture shares the genus. Of these copies, 734 were called a new lineage, the right rank, and the other 69 (8.6%) a new species in a known genus, and that genus was wrong in every case: *Rhizophagus* called *Sclerocystis* (69). With the LSU

reference intact, the number of copies with a wrong genus was 0. The genus test shows at the level of genera what the species test shows at the level of species: a copy whose clade has been removed is placed on the sister clade, here *Sclerocystis*, and named after the cultures that hold it. None of the 51 divergent copies was recognized as divergent in these rounds, because both divergent-copy tests need a related culture, and 40 of them also received the wrong genus.

3.7 The remaining errors are species complexes

The complex rule yields 4 complexes (Table 8): *Glomus chinense*, *Glomus macrocarpum* and *Glomus rugosae* (the VTX00199 group); *Diversispora aestuarii* with *Diversispora varaderana*; *Diversispora arenaria* with *Diversispora jakucsiae*; and *Rhizophagus irregularis* with *[Rhizoglomus] vesiculiferum*. Together they hold 9 of the 37 named species and 29.9% of the reference copies. One further pair, *Diversispora celata* with *Diversispora eburnea*, misses the LSU limit by 0.4 points and is not a complex.

Every one of the 53 false-known calls left after the ITS rule in the test species is a call to a member of the species' own complex. Without the ITS rule, 54 of the 78 errors are inside a complex and 24 (3.6% of the withheld copies) are not; with the LSU reference intact, 10 and 0 copies fall outside a complex without and with the rule. The ITS rule thus removes the errors that the markers could in principle have avoided, and what is left is the limit of the markers themselves. The complexes were derived from all cultures, including the withheld species, so this shows that the errors are confined to pairs that the reference already shows to be inseparable. It does not show that an undescribed relative of a complex member would be caught, which, by definition, it would not.

Table 8: Species complexes: pairs of named species whose nearest copies lie within the within-species spread on all three markers.

Species A	Species B	Complex	ITS	LSU	SSU
Diversispora aestuarii	*Diversispora varaderana*	C2	1.1	1.09	0.0
Diversispora arenaria	*Diversispora jakucsiae*	C3	3.48	0.83	0.07
Glomus chinense	*Glomus macrocarpum*	C1	5.87	2.18	0.27
Glomus chinense	*Glomus rugosae*	C1	4.95	1.36	0.2
Glomus macrocarpum	*Glomus rugosae*	C1	1.38	1.36	0.07
Rhizophagus irregularis	*[Rhizoglomus] vesiculiferum*	C4	3.25	2.44	0.34

Columns give the distance (%) between the nearest copies of the two species. The limits are the 99th percentile of the distance between cultures of one species: ITS 6.0%, LSU 3.0%, SSU 0.8%.

The complexes depend on the percentile. *Diversispora aestuarii* with *Diversispora varaderana*, *Diversispora arenaria* with *Diversispora jakucsiae*, and *Glomus macrocarpum* with *Glomus rugosae* are complexes at every percentile from the 95th to the 99.5th; *Glomus chinense* joins from the 97.5th, *Rhizophagus irregularis* with *[Rhizoglomus] vesiculiferum* from the 99th, and *Funneliformis coronatus* with *Funneliformis mosseae* at the 99.5th. Because the ITS distance rule uses the same percentile, a fair test moves both together (Table S5). At the 97.5th percentile the rule leaves the same 53 false-known calls as at the 99th, but the 18 copies of *Rhizophagus irregularis* called *[Rhizoglomus] vesiculiferum* now lie outside a complex. At the 95th the rule is stricter: false-known calls fall to 37 (5.5%), 5 of them outside a complex, all again *Rhizophagus irregularis* called *[Rhizoglomus] vesiculiferum*, at the cost of demoting 32 copies of known species in the culture test (4.2%) instead of 6. At every percentile, then, each remaining error is either inside a complex or in this one pair, and whether that pair counts as a complex is the only conclusion that depends on the choice.

3.8 Cultures from other collections

Before any test result was scored, the 850 copies of the 37 LSU-removal rounds were scored again with the test cultures present but excluded, and every call came back unchanged; adding the test sequences to the LSU alignment left the 903 reference placements unchanged.

Among the 102 ASVs of species the reference does not hold, 4 were called a known species with the species removed from the LSU reference (3.9%; 5.3% averaged over species and 3.8% over cultures; Table 9). These are exactly the 4 ASVs that the prediction had placed inside a reference species (*Gigaspora gigantea* KS302 called *Gigaspora rosea*); no false-known call fell outside the prediction. With the LSU reference intact, their LSU falls in its *Gigaspora gigantea* clade, which the reference's *Gigaspora rosea* cultures also occupy, and their ITS lies 0.5–2.0% from those cultures, so no marker separates the two. None of the 43 ASVs of genera absent from the reference was called known. Removing the species from the LSU reference changed no call: all 102 ASVs of absent species had the same call and placement either way, because their LSU fell in clades that no reference culture holds in both cases. These species are far from those of the reference, and the sister-species case behind the LSU-removal result (Section 3.6) occurred once, in KS302, where it was predicted.

Of the 164 ASVs of species the reference holds, 136 kept their species (82.9%), against 745 of 755 in the culture test. ASVs identical in ITS to a reference copy kept it in 40 of 43 cases, the others in 96 of 121 (79.3%). The ITS distance rule demoted 15 of them, whose ITS lies 6.1–9.6% from every reference culture, beyond its 6.0% limit: without the rule, 151 kept their species (92.1%), and the rule removed no false-known call (4 without it). The 8 ASVs called another species are *Funneliformis coronatus* MR101 called *Funneliformis mosseae*, as predicted (8 of 8): their ITS lies 0.0–0.6% and their SSU 0.0% from reference *Funneliformis mosseae* cultures, so the culture is most likely *Funneliformis mosseae* under another name. The prediction also sorted the known species: 135 of the 139 ASVs predicted inside their own species kept it, and 24 of the 25 predicted outside did not.

With the VT search nearly silent, the genus of absent species was hard to name: 44 of the 59 ASVs of absent species in genera the reference holds were called a new lineage, 9 were near a VT threshold, and the genus was given for 6, all correctly. With VT coverage counted over the part of the reference the ASV spans, the genus was given for 29, again all correctly, but two errors appeared: the 18 ASVs of *Racocetra gregaria* were given *Cetraspora*, through a VT the two genera share (VTX00041), while their LSU name was not carried; and one ASV, *Septoglomus constrictum* KS890 called *Septoglomus africanum* / *Septoglomus turnauae*, through a shared VT, with its ITS 1.1% from a reference culture. False-known calls then numbered 5 (4.9%), 1 of them outside the prediction.

The test also raised curation questions for the collections. Three culture codes carry different species in LongGloDB and in the Stefani et al. deposit (MT106, ON205A and BEG9). INVAM's name for both MT106 and ON205A is *Ambispora gerdemannii*, which Stefani et al. record and depart from (their Table S2) and which LongGloDB keeps for ON205A. In addition, the LongGloDB *Ambispora leptoticha* MX982A lies 7.8–9.6% in ITS from the reference's *Ambispora leptoticha*, ON205A, which LongGloDB names *Ambispora gerdemannii*.

4 Discussion

Linked copies turn the correspondence between VTs, SHs and LSU species from a guess into a measurement, and they show where each naming system misleads.

4.1 The three systems nest rather than match

SHs mostly sit inside LSU species, which mostly sit inside VTs. VT counts therefore underestimate species (at best, VTs separate 19 of 37), and SH counts overestimate them, because one

Table 9: Independent cultures: the rules as fixed, on 52 LongGloDB cultures from other collections.

Analysis	LSU reference	False known (of 102)	Share	Absent species: genus right/wrong/none (of 59)	Absent genera: genus given (of 43)	Known kept (of 164)
Main	intact	4 (4)	3.9%	6 / 0 / 53	0	136 (151)
	removed	4 (4)	3.9%	6 / 0 / 53	0	–
Second	intact	5 (5)	4.9%	29 / 0 / 30	18	139 (154)
	removed	5 (5)	4.9%	29 / 0 / 30	18	–

With the ITS distance rule; in brackets, the same without it. *LSU reference*: intact, or with each absent species removed in its own round. *False known*: ASVs of species the reference does not hold called a known species or species group. *Absent species*: ASVs of species the reference does not hold, in genera it holds. *Absent genera*: ASVs of genera it does not hold, so any genus given is wrong. *Known kept*: ASVs of species the reference holds whose call contains their species; LSU reference intact only. *Second*: the second analysis, fixed in advance, with VT coverage counted over the part of each MaarjAM reference that the ASV spans.

genome spans up to 6 SHs. Surveys that count SHs as taxa inflate AMF richness, and part of that inflation comes from variation within genomes. Grouping SHs by genome restores most of the species signal in our cultures, but the groups were learned from these same cultures, and whether they transfer to field sequences is untested. The 2.5-kb amplicons were introduced to bring the three regions into one dataset [24]; what was missing is a measured correspondence between the names, which linked copies supply.

4.2 Divergent copies are genomic; whether they work is open

Variation among the rDNA copies of an AMF genome has long been known [41, 49], yet it has been largely overlooked in ecological and taxonomic work [45], and a genome-based species classification has only recently been proposed [7]. What linked copies add is a measure of how this variation misleads each marker. Short-5.8S copies look like new lineages in the LSU and form their own SHs and VTs, several under placeholder names in UNITE. The short-5.8S test recognizes most of them from their sequence alone, but less often when the whole species is unknown, because no related culture with a 5.8S of the right length is then available for comparison. They are not artifacts: the same 5.8S deletion recurs across isolates and species, and on trees the short copies of different species group together. A long deletion that recurs should therefore be read as part of a known genome, not discarded as a singleton.

Being part of a genome does not make the short copies harmless. A species that carries them has two structural variants of the marker, so the short copies form ITS clusters of their own (18 at 98.5%, Table S7), and the ITS and LSU tell a different history from the SSU: on the LSU, the short copies of *Rhizophagus intraradices* and *Rhizophagus prolifer* sit next to *Sclerocystis* rather than with their own species. Because the class has a fixed signature, it can be flagged before clustering or tree building: in *Rhizophagus*, a 13-nt motif in the 5.8S marks every short copy and no other. Such a flag has to be read with the genus, because the same motif is the normal 5.8S of *Oehlia diaphana*, and it needs the 5.8S, which an ITS2 trimmed by ITSx no longer has. The motif was written from the copies it was tested on, so its error rate on other data is still unknown. Species are then best counted from main-type copies, and the short type analyzed as a locus of its own rather than mixed into a species tree.

Whether the short copies work is a separate question. Their SSU keeps its base pairing, and their LSU, although rich in substitutions, mostly keeps it too. The deletion shortens one hairpin of the 5.8S, so under the rule we fixed in advance the class counts as structurally degraded, as pseudogenes would be; yet the shortened stem is made of Watson–Crick pairs and is mostly closed by a UUCG loop, a combination known for stable hairpins [48]. DNA alone cannot tell

a transcribed and processed copy from a silent one. Sequencing the rRNA or its precursors from the same cultures would show whether the short type ends up in ribosomes. Same-length divergent copies are more ambiguous: co-dominant types of one genome in *Gigaspora*, second lineages in *Septoglomus*. They should be flagged, not explained away.

4.3 No identity threshold delivers species

The distances at each level of relatedness show why the three systems disagree (Figure S1). The thresholds of the three databases fall at different points on the scale of AMF variation. On the NS31–AM1 fragment, 97% identity would join almost every copy with a congener (98.7%) and some with another genus, mostly in *Cetraspora*, *Gigaspora* and *Dentiscutata*, which fits the coarse resolution of VTs (at most 19 of 37 species). On the full ITS, the 98.5% of SHs lies within the variation of a single culture, so one genome often spans several SHs. On the LSU, the 97% of V18 matches the spread within species, but 54.5% of copies have a congener inside it. On no region does one threshold separate species, in line with the absence of a barcode gap that Stefani et al. reported for each region [45], so moving a threshold only trades lumping against splitting. Taken together, the regions do better: on the Krüger fragment, 97% splits few genomes and, outside the complexes, joins almost no species, and the whole copy does the same at a higher identity. What no cut-off removes are the complexes and the short paralog. Combined names, with complexes reported as groups, avoid that trade-off where the markers allow it. The SSU, by contrast, behaves predictably: between cultures its distance was a median of 0.14 times the ITS distance, and no pair of cultures met the artifact test, so the test is conservative: an SSU that differs by at least half as much as the ITS points to an artifact, not to a genuine relative.

4.4 Novelty has a floor set by the markers

SSU, ITS and LSU together cannot separate the members of 4 complexes, 9 species in all: their nearest copies lie inside each other's within-species spread on every marker. All errors that remain after the ITS rule fall inside them at the 99th percentile used throughout. The claim holds at every percentile we tried except for one pair: with the rule and the complexes both at the 95th or 97.5th percentile, *Rhizophagus irregularis* and *[Rhizoglomus] vesiculiferum* are no longer a complex, and the *Rhizophagus irregularis* copies called *[Rhizoglomus] vesiculiferum* become errors outside one (Table S5). Whether these two names can be separated on rDNA is therefore the open question, not whether the floor exists. Within a complex, a species call is honest only as a group, and a call of novelty is impossible: an undescribed relative inside the spread of a known species cannot be told from that species on these regions. The alternative reading is that some of these names are not separate lineages, a question for taxonomists; *Diversispora aestuarii* and *Diversispora varaderana* differ by 0.0% in the SSU and 1.1% in the LSU. The rule is driven by the data and will change as cultures are added: one further pair (*Diversispora celata* with *Diversispora eburnea*) misses the LSU limit by 0.4 points.

4.5 Novelty depends on what the reference already knows

With only the cultures withheld, LSU placement looks like a strong detector of new species, but much of that strength comes from the withheld species still being in the LSU reference: its clade holds no culture, so its copies are flagged. Phylogenetic placement was proposed because matching against a database cannot place sequences of taxa that are absent from it [11, 9]. That holds at the level of clades, but the LSU-removal test shows its limit at the level of species: with the species removed, a copy is placed on its nearest named relative and then named after it, and with a genus removed, on the sister genus (69 *Rhizophagus* copies called *Sclerocystis*). An undescribed species is absent from every reference. Removing it from the LSU reference is therefore closer to the real situation than leaving it in, but not equal to it: MaarjAM and

UNITE were left intact, and many VTs are built from environmental sequences, so a species that is undescribed yet already sequenced in the field can still have a VT of its own. The VT accounted for the novelty of 18 of the 673 copies, so for a species never sequenced the false-known rate may be higher than reported, by up to about 2.7% of copies. The ITS distance rule does not depend on external databases and closes part of the gap, but not where the ITS itself overlaps. Pooled over the test species, false-known calls rose from 2.8% with the LSU reference intact to 11.6% with the species removed from it, and fell to 7.9% with the ITS rule.

4.6 Relation to earlier work on the same cultures

Stefani et al. [45] sequenced the copies used here (148 cultures of 44 species) and compared rDNA with three protein-coding genes (glomalin, H^+-ATPase and RPB1), sequenced for 143 cultures of 39 species. They found no barcode gap in any of the regions, so that no single distance threshold identifies AMF, and concluded that rDNA remains the practical choice because its primers work across all lineages. Our results agree where they overlap: the complexes are pairs whose nearest copies lie inside each other's spread on every rDNA region. Two of their observations follow from the short paralog. The rDNA of *Rhizophagus vesiculifer* (our *[Rhizoglomus] vesiculiferum*) and *Rhizophagus prolifer* was paraphyletic in their trees, nested in *Rhizophagus irregularis* and *Rhizophagus intraradices*; on our linked tree, these four species become clades once the divergent copies are set aside (Section 3.4). And their largest within-species distances on the Krüger fragment were in the species that carry the short type (*R. irregularis* 7.7%, *R. vesiculifer* 10.2%, *R. prolifer* and *R. intraradices* up to 11%) and in *Glomus chinense* and *Glomus rugosae* (7.4% and 8%), members of one of our complexes. Their distances are maximum pairwise distances over all copies, with every gap counted, and ours are distances to the nearest copy, so the numbers are not directly comparable. The questions differ. They asked whether a distance threshold identifies a species within one marker; we asked how the three naming systems relate and how far novelty can be called once divergent copies are accounted for. Their protein-coding sequences are public (glomalin PV808915–PV809055, RPB1 PV809056–PV809196, H^+-ATPase PV809197–PV809333) and could settle the complexes: if those genes separate *Diversispora aestuarii* from *Diversispora varaderana*, the complex is a limit of rDNA, and if not, it is a question of naming. We compared them (Table S6). At the thresholds Stefani et al. proposed, the three genes separate almost every pair of congeners that rDNA separates (42 of 44 on glomalin; 43 of 45 on RPB1; 43 of 45 on H^+-ATPase; 95–96%). Of the 6 pairs of species within complexes, 4 have protein-coding records for both species. Of these, 2 exceed the threshold on one gene each (*Diversispora aestuarii* and *Diversispora varaderana* on RPB1 (1.24% against 1.1%); *Rhizophagus irregularis* and *[Rhizoglomus] vesiculiferum* on glomalin (1.32% against 1.0%)), but none of the 12 comparisons exceeds the spread within the species themselves. The complexes are therefore not a limit of rDNA alone: on these genes, too, their members lie as close as cultures of one species, and whether they are separate species is a question for taxonomy rather than for the choice of marker. Joining the records through culture names also showed that the RPB1 and H^+-ATPase deposits rotate the species names of three cultures: FS16_2427A (*Diversispora epigaea* in the reference, *Septoglomus* sp. in GenBank); FS_9_2328B (*Septoglomus* sp. in the reference, *[Rhizoglomus] vesiculiferum* in GenBank); Gwalkeri_IVT132A (*[Rhizoglomus] vesiculiferum* in the reference, *Diversispora epigaea* in GenBank). Each of these records lies 0.0–0.9% from records of its culture's species, so the sequences are right and the names are not; we left the 6 records out.

4.7 Why the rules are not calibrated per species

A calibrated rule for calling ITS sequences new exists for fungi in general [37]. Here the limit of the ITS distance rule is itself a calibrated quantile, the 99th percentile of the distance between cultures of one species, and it behaves as set: it demoted 6 of the 755 copies of known species

(0.8%). The 99th percentile was fixed before the test species were run. The 95th would have cut false-known calls further (5.5% against 7.9% with the species removed from the LSU reference) at the cost of demoting 4.2% of the copies of known species instead of 0.8% (Table S5), a trade-off a user can set. The other rules, such as the demotion when a VT or LSU name is not carried, are not calibrated, because the reference holds too few cultures per species to fit a threshold for each, and because the traps peculiar to AMF, divergent copies and species complexes, have to be handled before a calibration can mean anything.

4.8 Independent cultures

On 52 cultures from three other laboratories, with the predictions fixed before the run, the rules called a species absent from the reference a known species only where the complex rule said they would: *Gigaspora gigantea* KS302, inside *Gigaspora rosea* on every marker. The test is easier than ours in one respect and harder in two. Its absent species are far from those of the reference, so it says little about the sister-species case, which occurred once. Its amplicons start inside the VT fragment, which leaves the VT search silent unless coverage is counted over the part a sequence spans, and a genus taken from the VT alone was then wrong when the true genus was absent from the reference (*Racocetra* given *Cetraspora*). And its cultures come from other collections, some of which lie further from ours in the ITS than our cultures of one species lie from each other: the ITS distance rule, fitted at the 99th percentile on one collection, demoted 15 of the 164 ASVs of known species and prevented no false-known call. Its limit should be fitted on more than one collection before the rule is used on field data.

4.9 Implications for surveys

Name at the level the markers support: report VT sets, SHs grouped by genome and LSU species clades, with species groups where the markers cannot decide, and do not expect any identity threshold to deliver species on its own. Screen for the short-5.8S class before counting taxa or building trees, and treat paralog SHs and VTs, such as SH1337353 and VTX00409, as known genomes. Base claims of novelty on evidence that does not come from the database that defines 'known', and read names against a stated taxonomy, because outdated names in a database look like errors of the marker [8]. Replication across independent libraries is planned for environmental data.

4.10 Limitations

Apart from the LongGloDB test, all results come from one collection of cultured species, and these cultures were used to develop the methods, so the results are development estimates. The LongGloDB test rests on 26 cultures of absent species, covers only distant novelty, and used amplicons on which the VT search as fixed is silent. The 99th percentile behind the ITS rule and the complexes was chosen after the distances in these cultures had been seen (Table S5 shows the effect of other choices), and the complexes were derived from all cultures, including the withheld species. Culture names are treated as truth, although some are doubtful: *Microdominikia litorea* 5274, some *Gigaspora rosea* cultures, and *Septoglomus* sp. C4 and 5070_5024A, whose LSU matches *Microviscospora peruviscosa*. Recurrence across culture labels can overstate independence, because one isolate may be held by several collections under different codes. MaarjAM and UNITE stay intact in every test; the LSU-removal test removes species from the LSU reference only. About a quarter of the V18 species labels with several records (51 in the placement audit) do not form exclusive clades on its backbone. The structure check reads DNA: a copy that keeps its base pairing need not be transcribed or end up in ribosomes, and a copy that loses a pair need not be silent. The distances at each level of relatedness are exploratory

and fix no rule. Replication on environmental libraries, and the copy-number model of amfsplit, are outstanding.

5 Conclusions

A call of novelty for an AMF sequence is only as good as the external references are complete, and it cannot be made below the resolution of the markers: with the withheld species removed from the LSU reference, some false-known calls remain, and every one of them lies in a species complex that SSU, ITS and LSU cannot separate. In cultures from other collections, the only species wrongly called known was the one the complex rule predicted. Linked long-read copies give an empirical bridge between the three naming systems and show where each one misleads. Divergent copies, which are genomic paralogs shared across species, must be recognized before taxa are counted, and species that the markers cannot separate, which protein-coding genes do not separate either, should be reported as groups.

Data and code availability

Sequences are GenBank PX207096–PX207274 and PX214440–PX215164. The culture reference (version 4, `181938d695f3df40`), the curation edits and their manifests, the scripts, and the test packages (withholding tests, LSU placement, and the LSU-removal test with its two protocol versions) are at `https://github.com/ayobi/amf-novelty-benchmark`, archived as release v0.1.0 at doi:10.5281/zenodo.23219691 [36], together with the LongGloDB test package (protocol, amendments, fixed predictions, runner and score tables). LongGloDB sequences are not redistributed; LongGloDB v1 is available from GloDB [14]. The copy checks (protocol and addendum with their checksums, scripts, alignments, trees, tables and figures) are in `benchmarks/copy_checks` of the same repository, and Supplementary Data S1 holds their alignments and trees.[TODO: archive a release that includes benchmarks/copy_checks and cite its DOI] External references: MaarjAM gDAT snapshot (sha256 `2b2c57e664e76347...`), UNITE 10.0 of 19 February 2025 [1] (UNITE+INSD sha256 `abe98f43e16c179f...`), Delavaux V18 AMF-only (sha256 `e6794cbbe166735a...`), and the Rfam models RF01960 and RF02543 (downloaded on 12 October 2026; sha256 `2cd46b2a3fe3f463...` and `63b122094f59d229...`) and RF00002 (sha256 `bc55434ebc3ef8c3...`), all three built in July 2024.

Author contributions

Aaron O'Brien conceived the study, developed the software, carried out the analyses and wrote the manuscript. César Marín reviewed the manuscript, in particular its treatment of AMF taxonomy and biology. Leho Tedersoo commented on the drafts and suggested further analyses.

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Competing interests

Aaron O'Brien develops ITSxRust [35], which is used in this study. The authors declare no other competing interests.

Use of generative AI

Anthropic's Claude and Microsoft Copilot (the versions current in 2026) were used in preparing this work: to draft and revise manuscript text and to write and debug analysis scripts. Aaron O'Brien revised the drafted text extensively, ran all analyses, and verified all reported values against the primary output; the authors are responsible for the content of the manuscript.

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Supplementary material

Methods S1. Barcode fragments

ITSxRust anchored each read with one nhmmer pass (HMMER 3 [13], ITSx fungal profile [4]) and delimited the SSU flank, ITS1, 5.8S, ITS2, full ITS and LSU flank. We used tolerant anchoring thresholds (E-value $\leq 10^{-3}$, anchor score ≥ 15), because anchoring only delimits regions. A panel of seven fragments (Table S1) was cut at the original primer sites, so that each fragment covers the region its database was built on. The panel used the primers of the SSU virtual taxa (NS31, AM1 and AML2 [44, 17, 29, 40]), the SSU V4 primers WANDA and AMV4.5NF/AMDGR [12, 42], the LSU primers LROR [5, 11] and FLR2 [47], the D2 primers FLR3 and FLR4 [15], and the outer pair AML1 and LSUmBr [29, 28] of the long amplicon [24], LSUmBr as the degenerate consensus of its five variants [43]. Primers were located with an IUPAC-aware infix alignment (edlib [52]) that tolerated edits up to 15% of the primer length; fragments were primer-trimmed, and coordinates are 1-based, inclusive and in rRNA-sense orientation. Every primer sequence was checked against published sources (`analysis/primer_check.tsv`). A primer was searched for only in the part of the read where it can lie (SSU primers before the ITS, LSU primers after it, plus 50 nt). A fragment is *open* when a primer site lies beyond the end of the copy, and *missing* when the primer is absent although the copy covers its expected position. Copies were also screened for repeats (concatemers) with the screen of amfsplit version 0.3.1, which will be described with the tool. No fragment crosses an ITSxRust anchor, and no copy was flagged as a repeat.

Methods S2. Culture reference

The names of the deposit are those of the culture tables of Stefani et al. (their Tables S2 and S3). Their 39 protein-coding species are phylogenetic species, which split *Septoglomus* sp. into three; their 44 rDNA species are not itemized, and we used the names of the deposit. A culture is an organism name plus a culture label. The reference was built by `freeze_ref.py` from the fragment tables and an edits file. Every edit stated first what the data had to look like (organism, label, copy count, VT, copy class, 5.8S length), and a failed expectation stopped the build before anything was written. Each version recorded input and output hashes and a fingerprint. The edits leading to version 4 (fingerprint `181938d695f3df40`) are listed in Table S2. Copy classes follow the 5.8S: copies whose 5.8S lies more than 3% from that of the culture's medoid copy were flagged, and the flagged copies were split into short-5.8S copies, at least 3 bp shorter than the culture's main type, and same-length copies. Where the medoid fell inside the short-5.8S group, the flags came out inverted, and the classes were swapped (Table S2).

Methods S3. Searches and candidate names

The VT search used the MaarjAM gDAT snapshot of July 2020 [40], the SH search UNITE release 10.0 of 19 February 2025 [34, 1], and the LSU search the Delavaux V18 database of May 2025 [11, 9, 10] (repository commit `f4a0014`). Searches used BLAST+ [6] with the settings in Table 1. Many V18 records carry terminal N padding, so coverage did not count reference Ns; without that correction, 95 of the 460 records would fail the coverage test even against themselves. Hits to any accession of the culture reference were dropped. The name of a culture in a system is the most common name of its main-type copies (ties went to the higher median identity). A copy's names depend only on its fragment, the database and the dropped accessions, not on culture names. The comparison of the names (the crosswalk) was run before the last two *Septoglomus* edits (Table S2), so its report on cultures was replayed on the final labels with the stored names of each copy (`crosswalk_replay.py`), after every stored name had been checked against evidence captured on the final reference: in all three systems and for all 903 copies, the

stored identity equaled that of the highest-scoring hit passing the threshold, and the best name was present at that identity.

Each copy received a set of candidate names instead of a single best hit. The anchor was the copy's best hit by bit score, and the set held every name within δ identity points of the anchor that also passed the identity threshold. Names that split the main-type copies of one culture at $\delta = 0$ were merged when the closest copies across the split were within 1% in the SSU flank; this limit keeps the two lineages of a mixed culture from being merged. δ was set to the smallest margin at which the main-type copies of at least 99% of cultures share a name; for VTs it was also at least the 95th percentile of the SSU differences within cultures. SHs and LSU names carry no such floor, because their variation within genomes is wide and uneven (95th percentile: ITS 8.6%, LROR–FLR2 3.9%), and the merged names carry that variation instead (Table S3).

Methods S4. LSU placement

The V18 references were realigned with MAFFT [21], and branch lengths and GTR+Γ_4 parameters were fitted on the published topology with RAxML-NG [26]. Queries were added with `-addfragments -keeplength` and placed with EPA-ng [3]. The compatible edges of a query carry at least 95% of its likelihood weight. Edges inside a maximal clade bearing a single V18 species label were collapsed into one name (the LSU species clade). Where a species label is not monophyletic on the backbone, its separate components stayed separate names, and edges that carry several species labels stayed individual names.

Methods S5. Combining the names and the decision steps

A name was linked to the species and genera of the training cultures whose candidate-name sets hold it. In each system a copy is *matched* (its name is carried) when its names belong to training cultures, *unmatched* (not carried) when it has a name but no training culture holds it, and *silent* when it has no name. Genera were intersected over the matched systems, and species over the matched systems whose names are linked to named species. The repository's codes for the decision steps are: likely artifact, A1; divergent copy, D1 (short 5.8S) and D2 (paralog name), with D3 for a same-length 5.8S that is only flagged; near a VT threshold, B1; known species or species group, K1 (all matched systems agree and neither the VT nor the LSU name is unmatched); new species in a known genus, N1 (VT unmatched), N2 (LSU unmatched), N3 (no species survives while a genus does) and N6 (the ITS distance rule); discordant, X1 (the matched systems share no genus); and new lineage, N4 (no system matched) and N5 (as N4, with the best VT also below the near-threshold band). The comparators of D1 are the main-type copies of the nearest training culture; the support of D2 is the number of other training cultures that reach the name only through short-5.8S copies. Divergent copies are named by the VT alone. A copy is scored against what the reference knows: main-type copies of species held by another training culture should receive a known species or species group that contains their species; main-type copies of withheld species should not receive a known species (otherwise a false-known call); and divergent copies should be recognized as divergent or, for same-length copies, at least not be called new.

Methods S6. Withholding tests and the choice of percentile

Every quantity derived from cultures was fitted on the training cultures of each round only: merged names, δ, the links from names to species, the comparators of D1, the support of D2, the pools of nearest copies and the limit of the ITS distance rule. The test with withheld cultures had 148 rounds, each removing one physical culture label together with all lineages sharing it; the test with withheld species had 50 rounds, one per named species with all its cultures removed plus one per culture not named to species. In these tests MaarjAM, UNITE and V18 stayed fixed.

In the LSU-removal test, V18 was realigned without the target's records, the model refitted and all copies placed again (Table S4). The second version of the protocol added the ITS distance rule (N6) and changed nothing else; *Glomus chinense* and *Glomus rugosae* informed that rule and are reported as development rounds. The test species (the 'first-outcome' rounds in the repository) are all other species: the 34 whose V18 records were removed, and *Ambispora callosa*, which V18 does not hold by label, reported apart. The same protocol had 15 genus rounds, in which every culture of a genus and the genus's V18 records were withheld together.

To test the choice of percentile, the ITS distance rule and the complexes were set together at the 95th and 97.5th percentiles (Table S5): the rule was re-applied to the stored results of the decision steps, with the limit of each round refitted on its training cultures (`percentile_sensitivity.py`). Percentiles follow Python's exclusive quantile method throughout, as in the complex rule (`species_groups.py`).

Methods S7. The independent test

The overlap of LongGloDB with the Stefani et al. cultures was established with `longglodb_overlap.py`. The protocol and its two amendments (`PROTOCOL_LGDB.md`) fixed the outcomes, conditions and analyses, and the predictions of the complex rule were locked by hash before any test sequence was searched, placed or scored. The frozen code of the second protocol version was used unchanged. All 151 reference cultures were used for training and every test culture was excluded, so no test sequence entered merges, margins, name links, comparators or the limit of the ITS distance rule. Test sequences were cut by the same fragment step as the deposited copies, and their VT and SH evidence was computed with the same filtering against the same releases. In condition A the test fragments were placed on the intact V18; in condition B each of the 19 absent species, all of which V18 holds by label, was removed from V18 in its own round, as in the LSU-removal test. The secondary analysis (amendment 2) measured VT coverage over the part of each MaarjAM reference that the query spans.

Methods S8. Trees, artifact and structure checks, and distances

The protocol (`benchmarks/copy_checks/PROTOCOL_COPY_CHECKS.md`) was locked on 11 October 2026 and the structure check (`ADDENDUM1_FUNCTION.md`) on 12 October 2026, each before its results were computed.

Trees. The copies of each order (Archaeosporales 30, Paraglomerales 28, Diversisporales 245 and Glomerales 600) were analyzed separately, because the ITS cannot be aligned across orders and every culture and species lies within one order. MAFFT v7.505 L-INS-i (`-localpair -maxiterate 1000`), without trimming; the linked alignment joins the three alignments with one partition per region. IQ-TREE 2.3.6 with ModelFinder (`-m MFP`; for the linked alignment, one model per partition with edge-linked proportional branch lengths), 1,000 ultrafast bootstraps (`-B 1000`), 1,000 SH-aLRT replicates (`-alrt 1000`), near-zero branches collapsed (`-czb`) and seed 20261011. Each tip is one distinct sequence and stands for every copy that carries it. A group is tested on a tree when it has at least two copies. It is *unresolved* when no branch separates it and none contradicts it (it sits in a polytomy, or shares an identical sequence with a copy of another group); a branch that only joins copies with the same sequence has no support value and never counts as strong. For a group that is not monophyletic, its *escapees* are its copies outside the largest clade made only of its copies. All 1,436 calls of monophyly were recomputed independently with Bio.Phylo, with the same result.

Artifact check. Each copy was aligned to the nearest main-type copy of its own culture (distance over the whole copy; never itself) by global alignment with free end gaps (Biopython `PairwiseAligner`; match 2, mismatch −3, gap opening −5 and extension −2, the BLASTN defaults). Gaps that touch either end of the alignment are trimming differences and were ignored. Each internal deletion was assigned to the region of the comparator (SSU flank, ITS1, 5.8S, ITS2,

LSU flank) that holds most of its bases. Two copies are of the same type when their full-ITS distance is at most 1.0%, and the two lineages of a mixed label count as one isolate. Sensitivity analyses fixed in advance: same type at 0.5% and 2.0%, and long deletions read as at least 10 nt in every region.

Structure check. Infernal 1.1.5 `cmalign`, with truncated alignments allowed. The groups were the short-5.8S copies, the same-length copies, the main-type copies with a long deletion in the SSU flank, 5.8S or LSU flank (*flagged main*), and all other main-type copies with a main-type culture-mate (the baseline, which set the 99th percentiles). The paired comparison set each divergent copy against the mean of the main-type copies of its culture (two-sided Wilcoxon signed-rank test). The 5.8S deletions of the short copies were located on the RF00002 consensus. The ITS2 was folded with ViennaRNA 2.7.2 `RNAfold` (default parameters, 37 °C): minimum free energy per nucleotide, and the base-pair distance between the structure of a copy and that of its comparator.

Distances. For pairs of cultures, each culture was represented by its medoid main-type copy, the copy with the smallest summed full-ITS distance to the other main-type copies of its culture. The within-species spread of the NS31–AM1 fragment (1.7%) was computed as for the three markers of the complex rule. The 28 Paraglomerales copies lack the NS31–AM1 fragment and were left out of that region.

Longer regions. The Krüger fragment was cut from each copy from the SSUmAf1 site [28] (found with at most one mismatch in all 903 copies) to the end of the copy at LSUmBr, so the last stretch of the SSUmAf–LSUmAr amplicon is missing; the whole copy runs from AML1 to LSUmBr. Distances, classes and spreads are as above. For a cut-off t, a main-type copy is split from its culture if it lies more than t from every other copy of its culture, and joined to another species if a copy of another species of its genus lies within t; the partners of the four complexes are counted both ways (Figure 3). Script: `stage3/copy_long_fragments.py`.

What the short type does. Short copies were counted by their 5.8S sequence: the short-5.8S class and every copy whose 5.8S is identical to one of it, which adds PX214776.1. All retained copies of the named species were clustered by single linkage within each order at 98.5% and 97% identity on the full ITS, the Krüger fragment and the whole copy, once with every copy and once with main-type copies only (Table S7). On the Glomerales tree of each marker, rooted on a *Funneliformis* tip, we recorded the clades made only of short copies, their support, the taxa they join and the main-type taxa of their sister group. Script: `stage3/copy_paralog_effects.py`.

A motif for the short type. The motif YTYGGRGARTATG covers the short-type hairpin of every short copy of *Rhizophagus* and *[Rhizoglomus]*. It was written from these copies and tested on the same copies, so it shows that such a flag is possible, not its error rate. We searched it in the 5.8S and in the whole copy of all 903 retained copies, and located the ITS2 primers fITS7 [19] and ITS3 [51] in the 5.8S, allowing up to two mismatches (ITS3 matched every *Rhizophagus* copy with one or two). Script: `stage3/short58_motif.py`; results in `benchmarks/copy_checks/short58_motif`. These analyses are exploratory and fix no rule.

Table S1: Barcode fragments cut from the 904 deposited rDNA copies (amfsplit v0.3.1; primer-trimmed fragments, primers found with up to 15% of their length in mismatches or gaps).

Fragment	Used by	Usable	Missing	Open	Median (bp)
SSU NS31–AM1	MaarjAM VTs	876	28 (3′)	0	508
SSU NS31–AML2	SSU studies	904	0	0	522
SSU WANDA–AML2	SSU V4	903	1 (5′)	0	500
SSU AMV4.5NF–AMDGR	SSU V4, short	904	0	0	216
LSU LROR–FLR2	LSU reference V18	904	0	0	735
LSU FLR3–FLR4 (D2)	LSU D2	904	0	26 (3′)	330
AML1–LSUmBr	full amplicon	0	904 (both)	0	–

Usable: complete plus open-ended fragments; missing: primer absent although the copy covers its position; open: primer site beyond the end of the copy. The 28 missing NS31–AM1 fragments are all the Paraglomerales copies, in which AM1 does not bind; the one missing WANDA fragment is the copy dropped from the reference (PX215117.1). The 26 open D2 fragments are copies that end before the FLR4 site. The outer amplicon primers are absent because the deposited copies are trimmed inside them. No fragment crosses an ITSxRust anchor, and none of the 904 copies is flagged as a concatemer.

Table S2: Curation edits from the deposited copies to version 4 of the culture reference.

Item	Edit	Evidence
PX215117.1 (*Gigaspora rosea*)	Dropped	SSU and ITS diverge at the same rate (4–6% and 5.6–7.6%): error or pseudogene
BR105_2435A	Second lineage split off as *Paraglomus* sp.	Three VTX00281 copies 28–33% from the culture's *Paraglomus brasilianum* copies in ITS
4375_IVT88A, ON205A	Labels merged	One culture filed under two organism names
T3_IVT34A, CC4_4A_IVT7A	Kept as *Rhizophagus irregularis*	ITS2 a median 3.3% from *Rhizophagus irregularis* and 6.4% from *[Rhizoglomus] vesiculiferum* (T3); nearest LSU reference *Rhizophagus irregularis*
SD5_IVT32, 212939_IVT38A	Main and divergent classes swapped	Medoid fell in the short-5.8S group
SO43BL, Blasz_17_2277A	Mixed cultures split; SO43BL named *Septoglomus africanum*	Reciprocal lineages; ITS and LSU agree on *Septoglomus africanum*

Table S4: LSU-removal test by species: false-known calls among the main-type copies of each withheld species.

Withheld species	Copies	LSU ref. intact		species removed	
		no rule	ITS rule	no rule	ITS rule
Acaulospora morrowiae	10	0	0	0	0
Ambispora callosa‡	20	20	0	20	0
Ambispora leptoticha	10	10	0	10	0
Cetraspora gilmorei	60	0	0	0	0
Dentiscutata heterogama	6	0	0	0	0
Diversispora aestuarii	20	1	1	20	20
Diversispora arenaria	4	0	0	4	4
Diversispora celata	10	0	0	0	0
Diversispora eburnea	15	0	0	0	0

Withheld species	Copies	LSU ref. intact		species removed	
		no rule	ITS rule	no rule	ITS rule
Diversispora epigaea	13	0	0	0	0
Diversispora jakucsiae	4	0	0	4	4
Diversispora spurca	18	0	0	0	0
Diversispora varaderana	31	8	8	3	3
Dominikia bonfanteae	6	0	0	0	0
Dominikia duoreactiva	9	0	0	0	0
Funneliformis caledonium	9	0	0	0	0
Funneliformis coronatus	24	0	0	0	0
Funneliformis geosporum	8	0	0	8	0
Funneliformis mosseae	130	0	0	0	0
Gigaspora rosea	51	0	0	0	0
Glomus chinense†	35	28	5	28	5
Glomus macrocarpum	7	0	0	4	4
Glomus rugosae†	75	4	4	75	62
Glomus tetrastratosum	21	0	0	0	0
Microdominikia litorea	12	0	0	0	0
Oehlia diaphana	29	0	0	0	0
Paraglomus brasilianum	8	0	0	0	0
Paraglomus laccatum	5	0	0	0	0
Rhizophagus intraradices	12	0	0	0	0
Rhizophagus irregularis	56	0	0	19	18
Rhizophagus prolifer	6	0	0	0	0
Rhizophagus silesianus	7	0	0	0	0
Sclerocystis sinuosa	18	0	0	0	0
Septoglomus africanum	6	0	0	6	0
Septoglomus turnauae	4	0	0	0	0
Septoglomus viscosum	21	0	0	0	0
[Rhizoglomus] vesiculiferum	23	0	0	0	0

False-known calls with the LSU reference (V18) intact or with the species removed from it, each without and with the ITS distance rule. † development species; ‡ species not in V18 by label (removal changes nothing).

Table S3: Candidate names on the culture reference (version 4).

System	Merged names	Margin δ	Consistent cultures	One name	Species resolved
VT	2	0.6	138/138	74%	16 of 37 (19 at $\delta = 0$)
SH	35	0	106/107	100%	32 of 37
LSU (nearest reference)	5	0	128/128	81%	27 of 37

A culture is consistent when the candidate names of its main-type copies share a name. *One name*: share of main-type copies with a single candidate name. A named species is resolved when no other named species shares any of its names. Merged names and species counts are learned and scored on the same cultures, so they are upper bounds.

Table S5: The ITS distance rule and the species complexes recomputed together at three percentiles.

Percentile	ITS limit (median)	Complexes (species)	Known demoted (of 755)	False known species withheld (of 803)	False known also LSU-removed (of 673)	Outside a complex
95th	4.4%	3 (6)	32 (4.2%)	9 (1.1%)	37 (5.5%)	5
97.5th	5.0%	3 (7)	22 (2.9%)	13 (1.6%)	53 (7.9%)	18
99th	6.0%	4 (9)	6 (0.8%)	18 (2.2%)	53 (7.9%)	0

The ITS distance rule is re-applied to the stored results of the decision steps at each percentile, with the limit fitted on the remaining cultures of each round, and the complexes are those of the complex rule at the same percentile. *Known demoted*: copies of known species that the rule demotes with one culture withheld. *Also LSU-removed*: the test species of the LSU-removal test. The 99th-percentile row reproduces the reported results exactly. Every error outside a complex is *Rhizophagus irregularis* called *[Rhizoglomus] vesiculiferum*.

Table S6: Protein-coding genes on the rDNA species complexes: smallest distance (%) between the two species, with the largest distance within either species in brackets.

Pair	Glomalin	RPB1	H^+-ATPase
Diversispora aestuarii / *Diversispora varaderana*	0.73 (0.0)	1.24 (1.73)	1.22 (0.0)
Diversispora arenaria / *Diversispora jakucsiae*	0.29 (–)	0.57 (–)	1.55 (–)
Glomus chinense / *Glomus rugosae*	0.73 (0.73)	0.33 (7.93)	1.00 (1.32)
Rhizophagus irregularis / *[Rhizoglomus] vesiculiferum*	1.32 (1.61)	0.81 (7.54)	1.18 (5.77)

Sequences of Stefani et al. (2025). Indicative species thresholds from that study: 1.0% / 1.1% / 1.7% (glomalin / RPB1 / H^+-ATPase). Distances are edlib infix edit distances including any intron, so they are comparable with each other but only approximately with those thresholds. Records are joined to species through the reference culture names; records whose GenBank name disagrees with the reference name of their culture are left out (6).

Table S7: Single-linkage clusters of the copies of the 37 named species, within each order, at two identity thresholds: all retained copies, and main-type copies only. Short only and same length only: clusters made only of short-5.8S or only of same-length divergent copies; several species: clusters holding copies of more than one species. Exploratory; no rule depends on it.

		Clusters		Clusters of all copies		
Region	Identity	all copies	main type	short only	same length only	several species
Full ITS	98.5%	240	220	18	2	2
Full ITS	97%	110	96	12	2	2
Krüger fragment	98.5%	110	95	13	2	1
Krüger fragment	97%	56	46	8	2	4
Whole copy	98.5%	71	58	11	2	3
Whole copy	97%	36	30	5	1	7

Supplementary Data S1

Alignments and maximum-likelihood trees of the culture copies, for each order and for the SSU flank, full ITS, LSU flank and linked copy: alignments (FASTA), trees (Newick, with SH-aLRT and ultrafast bootstrap supports on the branches, and tips labeled with accession, species, culture, copy class and number of copies), the copies each tip stands for, and the partitions of the linked alignment (`benchmarks/copy_checks/supplementary_data_S1`).

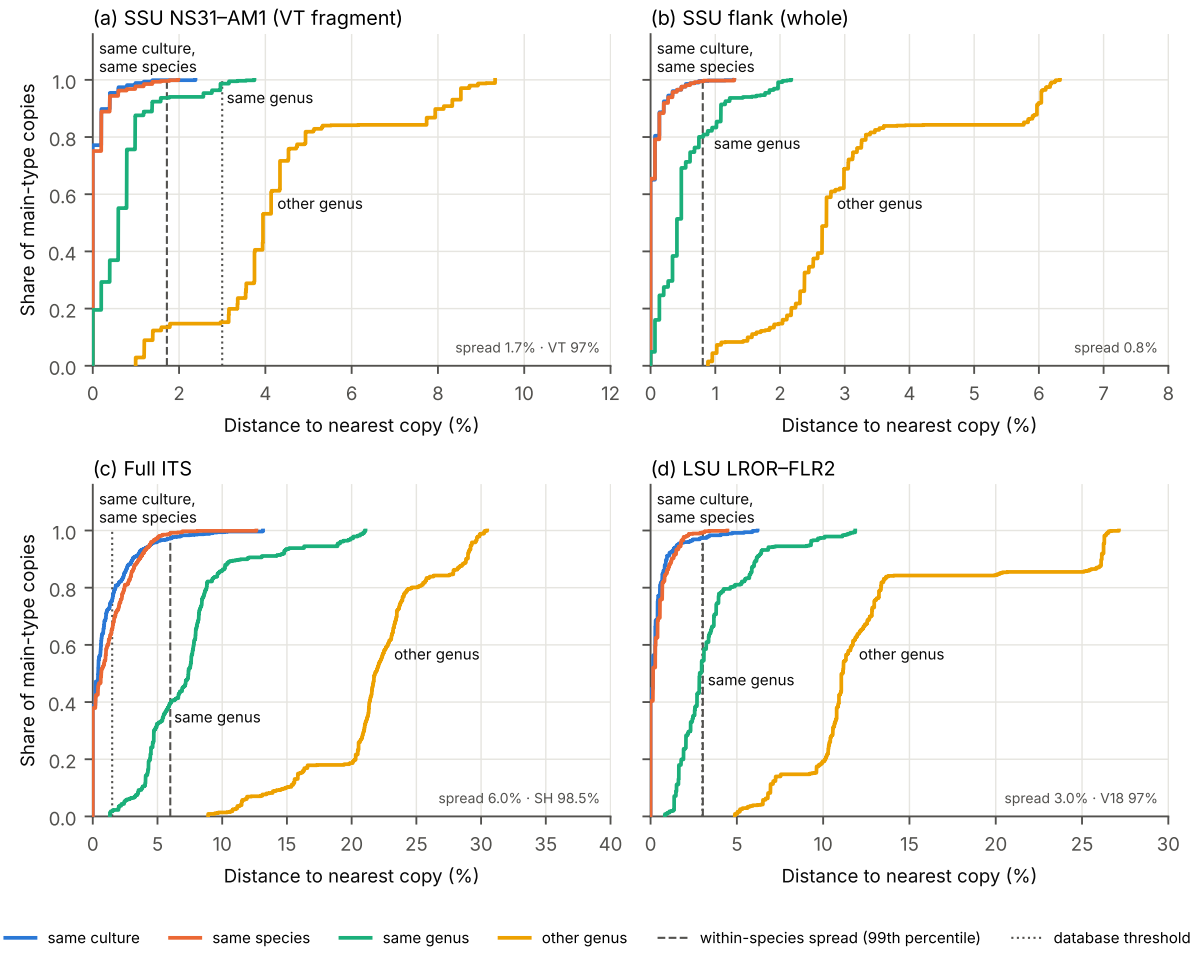


Figure S1: Distance from each main-type copy to its nearest main-type copy at four levels of relatedness: the same culture, another culture of the same species, another species of the same genus, and another genus of the same order. (a) The NS31–AM1 fragment, on which MaarjAM VTs are defined; (b) the whole SSU flank; (c) the full ITS; (d) the LSU fragment LROR–FLR2. Dashed lines, the within-species spread (the 99th percentile of the distance between cultures of one species); dotted lines, the identity thresholds of the databases on the fragments they apply to (VT 97%, SH 98.5%, V18 97%). The Paraglomerales copies lack the NS31–AM1 fragment.

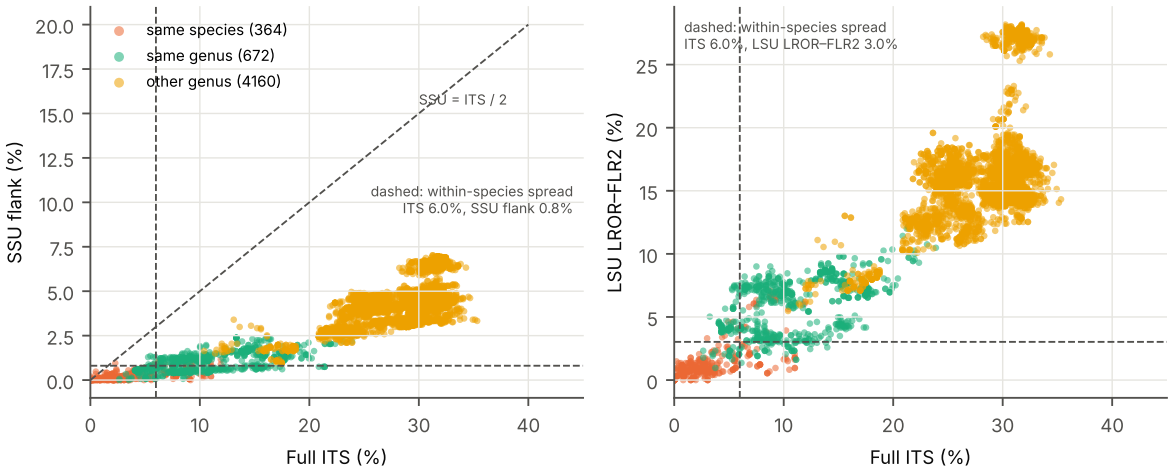


Figure S2: Distances between the medoid copies of pairs of cultures within an order: (a) SSU flank and (b) LSU LROR–FLR2 against full ITS, colored by relatedness (numbers of pairs in the legend). Dashed lines, the within-species spreads; in (a), the line SSU = ITS/2, above which the artifact test applies once the SSU differs by at least 1%.

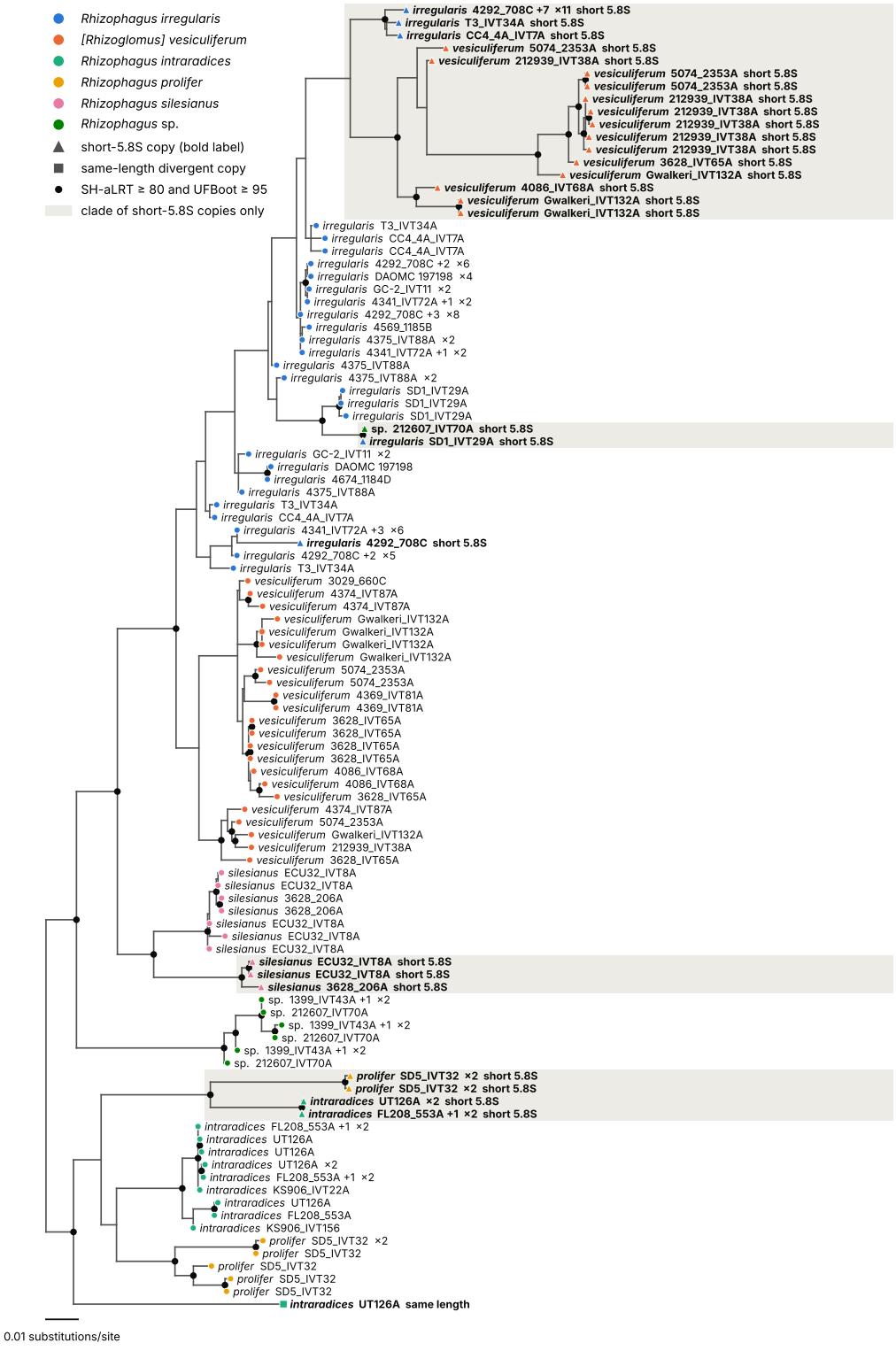


Figure S3: The uncollapsed *Rhizophagus* clade of Figure 2. Tips are distinct sequences, labeled with the species epithet and the culture; '$+n$', the same sequence also occurs in n further cultures; '$\times n$', the number of copies with that sequence. Triangles and bold labels, short-5.8S copies; square, the same-length copy; black dots, branches with SH-aLRT $\geq$ 80 and ultrafast bootstrap $\geq$ 95; shading, clades made only of short-5.8S copies.